

Quantum-Classical Physics-Informed Neural Networks for Solving Reservoir Seepage Equations

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Abstract: Solving partial differential equations (PDEs) for reservoir seepage is essential for optimizing oil and gas development and predicting production performance. Traditional numerical methods are hindered by mesh-dependent errors and high computational costs, while classical Physics-Informed Neural Networks (PINNs) struggle with parameter inefficiency, high-dimensional representation, and strong nonlinear fitting. To overcome these challenges, we adapt the Discrete Variable (DV)-Circuit Quantum-Classical Physics-Informed Neural Network (QCPINN) and, for the first time, apply it to four typical reservoir seepage models: the pressure diffusion equation for heterogeneous single-phase flow, the nonlinear Buckley-Leverett (BL) equation for simplified two-phase waterflooding, the convection-diffusion equation for compositional flow considering adsorption, and the fully coupled pressure-saturation two-phase oil-water seepage equation for heterogeneous reservoirs with exponential permeability distribution. The QCPINN integrates classical preprocessing/postprocessing networks with a DV quantum core, leveraging quantum superposition and entanglement to enhance high-dimensional feature mapping while embedding physical constraints to ensure solution consistency. We test three quantum circuit topologies (Cascade, Cross-mesh, Alternate) and demonstrate through four numerical experiments that QCPINNs achieve higher prediction accuracy than classical PINNs. Specifically, the Alternate topology outperforms others in heterogeneous single-phase flow, BL equation simulations and heterogeneous fully coupled pressure-saturation two-phase flow, while the Cascade topology excels in compositional flow with convection-dispersion-adsorption coupling. The Cross-mesh topology shows competitive early-stage convergence and accuracy across scenarios with balanced performance in coupled two-phase flow. Our work verifies the feasibility of QCPINN for reservoir engineering applications, bridging the gap between quantum computing research and industrial practice in oil and gas engineering.

Keywords: Quantum-Classical Physics-Informed Neural Network; Quantum Computing; Reservoir Numerical Simulation; Physics-Informed Neural Network; Flow in Porous Media; Discrete-Variable Quantum Circuit

1 Introduction

Solving partial differential equations (PDEs) is a core problem in the fields of science and engineering. The accurate solution of PDEs for reservoir seepage directly determines the reliability of oil and gas field development plan optimization and production performance prediction. Traditional numerical methods, such as the finite volume method (Rao et al., 2024), finite element method (Jackson et al., 2015), meshless method (Rao et al., 2022), and others (Wang et al., 2023; Rao et al., 2025a), rely on mesh or point cloud discretization. When dealing with strong reservoir heterogeneity, strong flow nonlinearity, and convection-dominated

problems, these methods are prone to mesh-dependent errors or numerical dispersion. Furthermore, the computational cost increases exponentially with the complexity of the system, making it difficult to meet the demand for efficient simulation of complex oil and gas reservoirs.

Physics-Informed Neural Networks (PINNs) embed physical laws into the neural network architecture (Raissi et al., 2019) and achieve PDE solving through a combination of data-driven and physics-constrained paradigms. They can ensure the physical consistency of solutions without requiring large amounts of labeled data, providing a new approach for reservoir forward simulation and inverse problem solving (Almajid & Abu-Al-Saud, 2022; Hanna et al., 2022). Recently, Physics-Informed Kolmogorov-Arnold Networks (PIKAN) have also been developed for reservoir flow simulation (Rao et al., 2025b; Kalesh et al., 2025). For instance, Rao et al. (2025b) applied PIKAN to single-phase seepage simulation of heterogeneous permeability based on a mixed pressure-velocity formulation and found that PIKAN outperformed PINNs in simulation performance. However, classical PINNs and PIKANs face three core limitations when addressing complex reservoir flow scenarios: low parameter efficiency, insufficient high-dimensional expression capability, and restricted fitting of strong nonlinearity. First, to capture the heterogeneity of permeability or the high-dimensional characteristics of governing equations, it is necessary to design deep fully connected networks or increase the number of neurons, leading to a surge in the scale of trainable parameters (usually ranging from thousands to tens of thousands). This not only increases the memory and time costs of training but also easily causes overfitting and unstable convergence. Second, when simulating the steep shock front of convection-dominated equations, the introduction of artificial viscosity terms (Fuks & Tchelepi, 2020) and the gradient propagation characteristics of classical activation functions (e.g., Tanh, ReLU) tend to result in gradient vanishing or local oscillations, making it difficult to balance accuracy and convergence speed. These limitations prevent classical PINNs from achieving both efficiency and accuracy in complex reservoir simulations, becoming a key bottleneck restricting their industrial application.

As a revolutionary paradigm to break through the bottlenecks of classical computing, quantum computing can trace its development back to the 1980s. Feynman (1982) proposed the core idea of quantum system simulation, and Deutsch (1985) confirmed the universal Turing machine property of quantum computers, laying a theoretical foundation for quantum computing. Subsequent landmark achievements, such as the Deutsch-Jozsa algorithm (1992), Shor's large integer factorization algorithm (1994), Grover's database search algorithm (1996), and the Harrow-Hassidim-Lloyd (HHL) linear system solving algorithm (2009), have theoretically verified the exponential acceleration advantages of quantum computing in specific problems. At the experimental level, Cirac and Zoller (1995) proposed the ion trap qubit scheme, Monroe et al. (1995) implemented the first CNOT quantum logic gate, and Chuang et al. (1998) verified the Deutsch-Jozsa algorithm based on nuclear magnetic resonance technology, promoting quantum computing from theory to experiment.

Currently, quantum computing is in the Noisy Intermediate-Scale Quantum (NISQ) era (Bharti et al., 2022), and variational quantum algorithms tailored for NISQ hardware have become a research focus (Cerezo et al., 2021), including the Variational Quantum Eigensolver (VQE) (Peruzzo et al., 2014; Tilly et al., 2022), Variational Quantum Linear Solver (VQLS) (Bravo-Prieto et al., 2023), Variational Quantum Classifier (VQC) (Havlíček et al., 2019; Jäger and Krems., 2023), and Variational Quantum Phase Estimation (VQPE) (Liu et al., 2024). These algorithms generate quantum states through Parameterized Quantum Circuits (PQC) and iteratively optimize parameters using classical optimizers. While avoiding noise accumulation in deep quantum circuits, they achieve approximate solutions to complex problems. In the field of PDE solving, VQLS has been attempted for solving Poisson equations (Ali and Kabel, 2023; Govindugari and Wong, 2024; Ayoub and Baeder, 2025), heat conduction equations (Liu et al., 2022), reservoir seepage equations (Rao., 2024a; Rao., 2024b), and others. By decomposing the coefficient matrix of the linear system (obtained from PDE numerical discretization) into a superposition of Pauli bases, VQLS computes the cost function using quantum

state preparation and Hadamard testing to obtain the solution of the linear system. Therefore, the application of VQLS to PDE computation requires prior numerical discretization of the PDE, rather than obtaining the PDE solution purely through quantum computing. Quantum neural networks (QNNs) based on variational quantum algorithms have also been used as pure machine learning methods (Tüysüz et al., 2021; Rao et al., 2024b); for example, Rao et al. (2024b) applied QNNs to predict carbon dioxide (CO₂) sequestration in saline aquifers. However, in general, due to the lack of physical information and the more challenging nonlinear processing compared to classical networks, the prediction performance of pure QNNs is relatively limited.

A natural idea is to develop pure Quantum Physics-Informed Neural Networks (QPINNs). For example, Trahan et al. (2024) applied QPINNs to spring-mass systems and Poisson equations, while Berger et al. (2025) applied QPINNs to Poisson, Burger's, and Navier-Stokes equations. Xiao et al. (2024) used QPINNs to solve forward and inverse problems of certain PDEs. Theoretically, however, pure quantum neural networks require more layers than hybrid quantum-classical neural networks, which significantly increases the hardware threshold for physical implementation of the network and makes it more susceptible to noise interference. Moreover, hybrid quantum-classical neural networks can leverage the advantage that classical networks achieve nonlinear complexity at a much lower cost than quantum networks, thereby enhancing the nonlinear fitting capability of the neural network.

Thus, hybrid Quantum-Classical Physics-Informed Neural Networks (QCPINNs) have been further developed. For instance, Sedykh et al. (2024) proposed a QCPINN for simulating laminar fluid flow in 3D Y-shaped mixers, and Dehaghani et al. (2024) applied QCPINNs to solve quantum optimal control problems. Fernandez et al. (2025) integrated a quantum layer into a classical PINN to form a QCPINN and conducted preliminary tests on its computational performance for wave equations. Trahan et al. (2024) and Leong et al. (2025) applied QCPINNs to 1D Burger's equations and 2D inviscid compressible flows, respectively. Farea et al. (2025) applied QCPINNs to Helmholtz equations, wave equations, Klein-Gordon equations, and other PDEs. Through numerical experiments, both Trahan et al. (2024) and Farea et al. (2025) found that compared with classical PINNs, QCPINNs can construct high-dimensional feature spaces using the properties of quantum superposition and entanglement. This enables PDE solving with far fewer parameters than classical PINNs, significantly improving parameter efficiency. QCPINNs provide a breakthrough solution to the core limitations of classical PINNs: through a hybrid architecture of classical preprocessing, quantum core computing, and classical postprocessing, they achieve deep integration of quantum advantages and physical constraints.

As a newly developed method in the past two years, QCPINNs have so far been applied primarily to fundamental physical PDEs and have not been extended to the field of reservoir engineering. Therefore, this study for the first time applies QCPINN technology to reservoir engineering, systematically simulating four typical reservoir seepage models: the pressure diffusion equation for single-phase reservoir flow with heterogeneous permeability, the Buckley-Leverett (BL) equation for simplified two-phase waterflooding, and the convection-diffusion equation for component concentration considering adsorption in compositional flow, and the fully coupled pressure-saturation two-phase seepage equation for heterogeneous reservoirs with exponential permeability distribution. It accurately captures core features such as permeability heterogeneity, flow nonlinearity, and convection dominance, providing an early test case for the future development of reservoir numerical simulators or machine learning surrogate models based on quantum computing. This work promotes the translation of quantum computing from basic physics research to industrial application in oil and gas engineering.

2 Typical Partial Differential Equations in Reservoir Numerical Simulation

This section briefly presents the corresponding typical partial differential equations: the pressure diffusion equation for heterogeneous permeability in single-phase flow, the nonlinear Buckley-Leverett (BL) equation

for two-phase flow, and the convection-diffusion equation for component concentration considering adsorption effects.

2.1 Pressure Diffusion Equation with Heterogeneous Permeability in Single-Phase Flow

The core of single-phase flow (e.g., pure oil-phase flow in the early stage of reservoir development) lies in the diffusion process of the pressure field in a medium with heterogeneous permeability. Its governing equation is derived based on Darcy's law and the law of mass conservation. For a two-dimensional (2D) reservoir domain $\mathbf{x} = (x, y)$ and time domain $t \in [0, T]$, the general form of the pressure diffusion equation for heterogeneous single-phase flow is:

$$\frac{\partial}{\partial t}(\phi(\mathbf{x})\rho) = \nabla \cdot \left(\frac{k(\mathbf{x})}{\mu} \rho \nabla p \right) + q_s(\mathbf{x}, t), \quad (1)$$

where $p(\mathbf{x}, t)$ denotes reservoir pressure, the unknown function to be solved; $\phi(\mathbf{x})$ is porosity; $k(\mathbf{x})$ represents the absolute permeability of the rock; ρ is fluid density, assumed constant under weakly compressible conditions; μ denotes fluid viscosity; $q_s(\mathbf{x}, t)$ is the source/sink term (negative for production wells where fluid flows out, positive for injection wells where fluid flows in). Under steady-state and weakly compressible fluid conditions, Eq. (1) can be further simplified to an elliptic equation for pressure:

$$\nabla \cdot \left(\frac{k(\mathbf{x})}{\mu} \nabla p \right) + \frac{q_s(\mathbf{x})}{\rho} = 0, \quad (2)$$

Boundary conditions include Dirichlet boundaries (constant pressure boundaries, such as edge water or bottom water boundaries) and Neumann boundaries (constant flux boundaries, such as closed faults or impermeable boundaries), which satisfy $p(\mathbf{x}) = p_b$, $(\mathbf{x} \in \Gamma_D, t \in [0, T])$ and $\nabla p(\mathbf{x}) \cdot \mathbf{n} = 0$, $(\mathbf{x} \in \Gamma_N, t \in [0, T])$, respectively. Here, $\Gamma_D \subset \partial\Omega$ and $\Gamma_N \subset \partial\Omega$ are the constant pressure and constant flux boundaries, p_b is the boundary pressure, and $\mathbf{n}$ is the boundary normal vector. The spatial heterogeneity of permeability $k(\mathbf{x})$ is the key factor leading to the nonlinearity of pressure distribution.

2.2 Nonlinear Buckley-Leverett (BL) Equation in Two-Phase Flow

The waterflooding process is a typical two-phase flow, and the BL equation describes the nonlinear convection process of saturation through simplified mass conservation relationships, requiring special handling of the steep gradient at the shock front. Under the one-dimensional (1D) flow assumption (along the main streamline direction x), ignoring gravity and capillary forces, the conservative form of the BL equation is:

$$\frac{\partial S_w}{\partial t} + v \frac{\partial f_w(S_w)}{\partial x} = 0, \quad (3)$$

where $S_w(x, t)$ is water saturation; v denotes the total liquid flow velocity; $f_w(S_w)$ is the water cut, representing the ratio of water flow rate to total flow rate, typically a strongly nonlinear function of saturation with the expression:

$$f_w(S_w) = \frac{1}{1 + \frac{\mu_w k_{ro}(S_w)}{\mu_o k_{rw}(S_w)}}, \quad (4)$$

where $k_{rw}(S_w)$ and $k_{ro}(S_w)$ are the relative permeabilities of the water and oil phases, respectively, usually described by the Corey model:

$$k_{rw}(S_w) = \begin{cases} 0 & S_w \leq S_{wc} \\ \left(\frac{S_w - S_{wc}}{1 - S_{wc} - S_{or}} \right)^{n_w} & S_{wc} < S_w < 1 - S_{or} \\ 1 & S_w \geq 1 - S_{or} \end{cases}, \quad k_{ro}(S_w) = \begin{cases} 1 & S_w \leq S_{wc} \\ \left(\frac{1 - S_w - S_{or}}{1 - S_{wc} - S_{or}} \right)^{n_o} & S_{wc} < S_w < 1 - S_{or} \\ 0 & S_w \geq 1 - S_{or} \end{cases}, \quad (5)$$

in which, S_{wc} is the irreducible water saturation, S_{or} is the residual oil saturation, and n_w , n_o are Corey exponents, all determined by core experiments.

The initial condition of the BL equation is generally that the reservoir is saturated with pure oil before waterflooding, and the water saturation equals the irreducible water saturation: $S_w(x, 0) = S_{wc}$ for $(x \in [0, L])$, where L is the length of the main streamline in the reservoir. The boundary condition generally adopts a constant saturation boundary at the injection end ($x = 0$): $S_w(0, t) = 1 - S_{or}$, $(t \in [0, T])$. The core challenge of the BL equation is the shock front induced by nonlinear convection: when the fractional flow function $f_w(S_w)$ is convex, the saturation distribution forms a steep discontinuous front whose position advances with time.

2.3 Convection-Diffusion Equation for Concentration in Compositional Flow

Chemical flooding (e.g., polymer flooding) or CO₂ sequestration processes in reservoir development involve convection, diffusion, and adsorption of multicomponent substances on rock surfaces. Taking a 2D reservoir domain $\Omega \subset \mathbb{R}^2$ as an example, the convection-diffusion equation for the concentration of a single component (e.g., polymer) is:

$$\phi \frac{\partial c}{\partial t} + \rho_b \frac{\partial q}{\partial t} = \nabla \cdot (D \nabla c) - \nabla \cdot (\mathbf{v} c), \quad (6)$$

where $c = c(\mathbf{x}, t)$ denotes the component concentration; t represents time; ϕ is the porosity; ρ_b stands for the rock density; q denotes the mass of solute adsorbed per unit mass of rock. It satisfies the linear adsorption law (Henry's law) with the concentration c , i.e., $q = K_d c$, where K_d is the distribution coefficient reflecting the rock's adsorption capacity for the component; D represents the diffusion coefficient of the component; $\mathbf{v} = (v_x, v_y)$ denotes the seepage velocity distribution.

The initial condition of the multicomponent convection-diffusion equation is that no injected components are present in the reservoir at the early development stage, with a concentration of zero: $c(\mathbf{x}, 0) = 0, (\mathbf{x} \in \Omega)$. Boundary conditions generally include: a constant concentration injection at the injection boundary $\mathbf{x} \in \Gamma_{in}$ denoted by $c(\mathbf{x}, t) = c_{inj}$, $(t \in [0, T])$; a convection-dominated free outflow at the production boundary $\mathbf{x} \in \Gamma_{out}$ denoted by $\nabla c(\mathbf{x}, t) \cdot \mathbf{n} = 0$, $(t \in [0, T])$; and a closed boundary $\mathbf{x} \in \Gamma_{close}$ denoted by $\nabla c(\mathbf{x}, t) \cdot \mathbf{n} = 0$ and $c(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) \cdot \mathbf{n} = 0$, $(t \in [0, T])$. In reservoir seepage, the convection velocity is much higher than the diffusion velocity, resulting in convection-dominated equations, sharp concentration fronts, and a high tendency for numerical oscillations.

3 Discrete Variable (DV)-Circuit QCPINN Method

The core idea of the DV-Circuit QCPINN (Farea et al., 2025) is to leverage classical networks for input dimension adaptation and feature preprocessing, utilize the superposition/entanglement properties of DV quantum circuits to achieve high-dimensional feature mapping, then employ classical networks again to decode quantum outputs and calculate physical losses, and ultimately optimize both classical and quantum parameters through backpropagation.

Targeting the characteristics of reservoir PDEs, this study adapts this framework to ensure stability and accuracy under scenarios of heterogeneity, strong nonlinearity, and convection dominance. As illustrated in Fig. 1, this framework comprises four components: a classical preprocessor, a DV quantum core, a classical postprocessor, and a physical loss calculation unit. Specifically, the preprocessor maps reservoir inputs (spatiotemporal coordinate parameters of pressure, saturation, or concentration) to quantum input dimensions; the quantum core generates quantum states via parameterized DV circuits and performs measurements; the postprocessor decodes quantum measurement results into physical variables (pressure, saturation, or concentration); and the loss unit computes PDE residual loss as well as boundary/initial condition losses, with all parameters optimized through backpropagation.

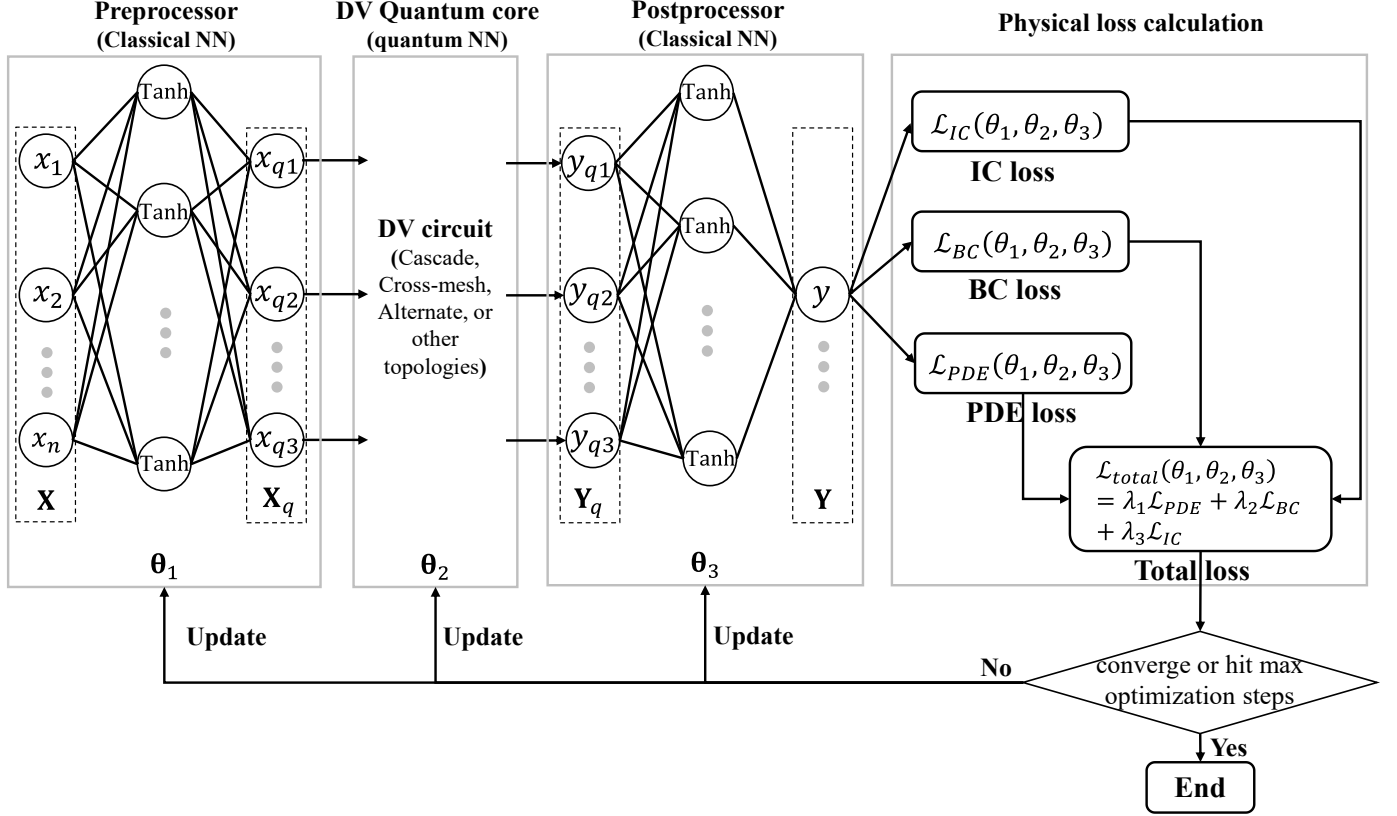


Fig. 1 Schematic diagram of the QCPINN workflow proposed in this study.

Next, we elaborate on the technical details of each component.

3.1 Classical preprocessor

The core functions of the classical preprocessor are input dimension adaptation and feature normalization. It converts inputs of reservoir seepage PDEs (e.g., spatiotemporal coordinates x, y, t for 2D pressure equations) into low-dimensional quantum inputs acceptable to quantum circuits, while reducing sensitivity to parameter initialization. Specifically, it is designed as a fully connected network with a single hidden layer (i.e., one input layer, one hidden layer, and one output layer): the input is the feature vector $\mathbf{X} \in \mathbb{R}^{d_{in}}$ of the reservoir PDE (e.g., for 2D pressure equations, including x, y), and the number of neurons in the hidden layer is d_h ($d_h = 50$ in the numerical examples of this paper). The linear transformation from the input layer to this hidden layer is given by $\mathbf{h}_1 = \mathbf{W}_1 \mathbf{X} + \mathbf{b}_1$, where $\mathbf{W}_1 \in \mathbb{R}^{d_h \times d_{in}}$ is the weight matrix, $\mathbf{b}_1 \in \mathbb{R}^{d_h}$ is the bias vector, and Xavier initialization is adopted (to avoid gradient vanishing). A Tanh activation function is introduced in the hidden layer to enhance the nonlinear expression capability of the network. Subsequently, the output layer data $\mathbf{X}_q \in \mathbb{R}^{d_q}$ is obtained through full connection between the hidden layer and the output layer (where d_q is the number of qubits used in the quantum circuit), serving as the input to the subsequent quantum circuit,

i.e., $\mathbf{X}_q = W_2 \text{Tanh}(\mathbf{h}_1) + b_2$. Thus, Eq. (7) presents the computation result of the preprocessor for the feature vector $\mathbf{X}$ of the reservoir PDE. For convenience of description, the set of parameters W_1 , W_2 , b_1 , b_2 involved in the classical preprocessor is denoted as θ_1 .

$$\mathbf{X}_q = W_2 \text{Tanh}(W_1 \mathbf{X} + b_1) + b_2 = f_1(\mathbf{X}, \theta_1), \quad (7)$$

where f_1 represents the mapping from the input $\mathbf{X}$ of the preprocessing network and the preprocessing network parameters θ_1 to the input $\mathbf{X}_q$ of the quantum network.

3.2 DV Quantum Core

The DV quantum core is the computational engine of QCPINN, responsible for encoding classical preprocessed features into quantum states, achieving high-dimensional feature mapping via Parameterized Quantum Circuits (PQL), and ultimately obtaining classical outputs through measurements.

3.2.1 Quantum State Encoding

This paper adopts angle embedding to map classical features to quantum states, in which each classical feature independently controls the rotation angle of one qubit, ensuring stable gradient propagation. For quantum input $\mathbf{X}_q = [x_{q1}, x_{q2}, \dots, x_{qd_q}]$, each element x_{qj} is treated as a rotation angle. We sequentially apply an RX rotation gate to each qubit j to implement feature encoding, with the unitary transformation of the RX gate defined as:

$$R_X(x_{qj}) = \begin{pmatrix} \cos(x_{qj}/2) & -i \sin(x_{qj}/2) \\ -i \sin(x_{qj}/2) & \cos(x_{qj}/2) \end{pmatrix}. \quad (8)$$

The initial quantum state of the system is the d_q -qubit ground state $|\psi_0\rangle = |0\rangle^{\otimes d_q}$. After applying the RX gates to all qubits, the encoded quantum state is:

$$|\psi_{\text{enc}}\rangle = \bigotimes_{j=1}^{d_q} R_X(x_{qj}) |0\rangle^{\otimes d_q}. \quad (9)$$

Through local rotation operations, angle embedding directly encodes classical features into the phase of qubits, avoiding coupling errors caused by global normalization.

3.2.2 Parameterized Quantum Circuit (PQC) Topologies

The topology of the quantum circuit determines the entanglement pattern between qubits and directly affects the feature mapping capability. In each subsequent numerical example, three representative parameterized quantum circuit topologies (Cascade, Cross-mesh, and Alternate) are tested. Specifically, the Cascade topology traces its origin to the work of Dallaire-Demers and Killoran (2018); the Cross-mesh topology is derived from the universal quantum circuit framework introduced by Sousa and Ramos (2006); the Alternate topology builds on the device-tailored gate sequence designed by Johnson et al. (2017). These three architectures are systematically collated and validated as typical designs in Sim et al., (2019). Their mathematical formulations and structural details are as follows:

(1) Cascade Topology

Fig. 2(a) shows a single-layer Cascade topology for d_q qubits, adopting a ring connection. Each layer of the circuit consists of two sub-layers: a single-qubit rotation layer and a two-qubit entanglement layer.

The single-qubit rotation layer applies $R_X(\alpha_j)R_Z(\beta_j)R_X(\gamma_j)$ to each qubit j , where α_j , β_j , γ_j are trainable parameters. The unitary transformation of the RZ gate is:

$$R_Z(\beta_j) = \begin{pmatrix} e^{-i\beta_j/2} & 0 \\ 0 & e^{i\beta_j/2} \end{pmatrix}. \quad (10)$$

The entanglement layer applies Controlled-RX (CRX) gates along the ring connection. The CRX gate with control qubit k and target qubit m is defined as:

$$\text{CRX}(\delta_{km}) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes R_X(\delta_{km}), \quad (11)$$

where δ_{km} is the trainable entanglement parameter, and I is the 2×2 identity matrix.

For d_q qubits and L layers, the total unitary transformation of the Cascade PQC is:

$$U_{\text{Cascade}}(\boldsymbol{\theta}_2) = \prod_{l=1}^L \left[\prod_{(k,m) \in \text{ring}} \text{CRX}(\delta_{l,km}) \prod_{j=1}^{d_q} R_X(\alpha_{l,j}) R_Z(\beta_{l,j}) R_X(\gamma_{l,j}) \right], \quad (12)$$

in which, for ease of description, the set of all parameters involved in the quantum circuits (including encoding rotation angles and PQC trainable parameters) is denoted as $\boldsymbol{\theta}_2$, and $\text{ring} = \{(0,1), (1,2), \dots, (d_q-2, d_q-1), (d_q-1, 0)\}$.

The PQC in Fig. 2(a) shows $d_q = 3$ qubits and $L=1$ layer (a constraint of NISQ-era hardware), the circuit depth is $(d_q + 2)L = 5$, and the number of parameters is $3 \times d_q \times L = 9$, including $2d_q \times L = 6$ single-qubit parameters and $d_q \times L = 3$ entanglement parameters, far fewer than the thousands of parameters in classical PINNs.

(2) Cross-mesh topology

Fig. 2(b) depicts a single-layer Cross-mesh topology for d_q qubits, featuring a fully connected entanglement structure (each qubit is entangled with all other qubits). The circuit consists of three sub-layers in sequence: a pre-single-qubit rotation layer, a fully connected entanglement layer, and a post-single-qubit rotation layer:

The pre-single-qubit rotation layer applies $R_X(\alpha_j) R_Z(\beta_j)$ to each qubit j , where α_j and β_j are trainable parameters.

The entanglement layer applies Controlled-RZ (CRZ) gates to all qubit pairs (k, m) where $k \neq m$. The CRZ gate with control qubit k and target qubit m is defined as:

$$\text{CRZ}(\delta_{km}) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes R_Z(\delta_{km}). \quad (13)$$

The post-single-qubit rotation layer applies $R_X(\gamma_j) R_Z(\eta_j)$ to each qubit j , where γ_j and η_j are additional trainable parameters.

The total unitary transformation of the Cross-mesh PQC is:

$$U_{\text{Cross-mesh}}(\boldsymbol{\theta}_2) = \prod_{l=1}^L \left[\prod_{j=1}^{d_q} R_X(\gamma_{l,j}) R_Z(\eta_{l,j}) \prod_{1 \leq k \neq m \leq d_q} \text{CRZ}(\delta_{l,km}) \prod_{j=1}^{d_q} R_X(\alpha_{l,j}) R_Z(\beta_{l,j}) \right]. \quad (14)$$

The PQC in Fig. 2(b) shows $d_q = 3$ and $L=1$, the circuit depth is $(d_q^2 - d_q + 4)L = 10$ layers, and the number of parameters is $(d_q^2 + 3d_q)L = 18$, including $4d_q \times L = 12$ single-qubit parameters and $d_q(d_q - 1)L = 6$ entanglement parameters.

(3) Alternate topology

Fig. 2(c) illustrate the Alternate topology for d_q qubits, which employs a sophisticated nearest-neighbor entanglement strategy with sequential gate operations. Different from the basic nearest-neighbor connection, the topology implements a multi-step gate sequence per layer to enhance entanglement diversity, specifically consisting of six sequential steps:

Step 1: Applies a RY rotation gate to all qubits simultaneously, where $RY(\alpha_j)$ (with α_j as trainable parameters) is the RY gate for qubit j . Its unitary transformation is defined as:

$$R_Y(\alpha_j) = \begin{pmatrix} \cos(\alpha_j/2) & -\sin(\alpha_j/2) \\ \sin(\alpha_j/2) & \cos(\alpha_j/2) \end{pmatrix}. \quad (15)$$

Step 2: Performs CNOT gates alternately on adjacent qubit pairs. Specifically, for even-indexed pairs $(0,1), (2,3), \dots$ (or odd-indexed pairs, depending on layer configuration), a CNOT gate is applied with the first qubit as control and the second as target. The CNOT gate is defined as:

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X. \quad (16)$$

where X is the Pauli-X gate, $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Step 3: Applies an RZ rotation gate ($RZ(\beta_j)$, β_j as trainable parameters) to all qubits that served as control or target qubits in Step 2. The unitary transformation of the RZ gate is consistent with Eq. (17).

Step 4: For the adjacent qubit pairs that did not receive CNOT gates in Step 2 (denoted as "remaining pairs"), apply an RY rotation gate ($RY(\gamma_k)$, γ_k as trainable parameters) to the qubits in these pairs whose last operation was not an RY gate (i.e., the qubits in remaining pairs that were not involved in Step 2's CNOT gates).

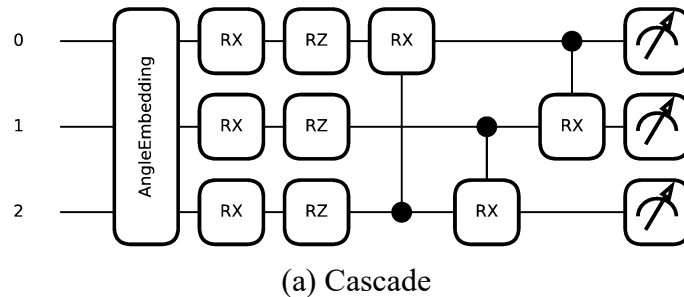
Step 5: Deploy CNOT gates on all remaining pairs (those without CNOT in Step 2). For each remaining pair (j,k) (where j is the upper qubit and k is the adjacent lower qubit), the CNOT gate is applied with the upper qubit as control and the lower qubit as target (consistent with the control-target order in Step 2), with the CNOT gate form consistent with Eq. (22).

Step 6: Applies an RZ rotation gate ($RZ(\delta_l)$, δ_l as trainable parameters) to all qubits whose last operation was not an RZ gate, ensuring each qubit ends with an RZ rotation to adjust the phase.

The total unitary transformation of the Alternate PQC for L layers is:

$$U_{\text{Alternate}}(\theta_2) = \prod_{l=1}^L \left[\prod_{\substack{i=1 \\ \text{last op} \neq RZ}}^n RZ(\delta_{l,i}) \prod_{\text{remaining pairs}} \text{CNOT}_{l,(j,k)} \prod_{\substack{k \in \text{remaining pairs} \\ \text{last op} \neq RY}}^n RY(\gamma_{l,k}) \right] \prod_{\substack{j \in \text{CNOT-involved} \\ \text{Step 2}}}^n RZ(\beta_{l,j}) \prod_{\text{alternate pairs}} \text{CNOT}_{l,(j,k)} \prod_{i=1}^n RY(\alpha_{l,i}). \quad (17)$$

The PQC in Fig. 2(c) shows $d_q=3$ qubits and $L=1$ layer, the circuit depth is $6L=6$ layers. When d_q is odd, the total number of parameters is $4(d_q - 1)L = 8$; when d_q is even, the number of parameters is $4d_qL$. All these parameters belong to single-qubit gates. This design not only captures local dependencies in physical systems through alternating rotation and entanglement operations but also reduces resource consumption and noise impacts of quantum hardware via its concise connection scheme.



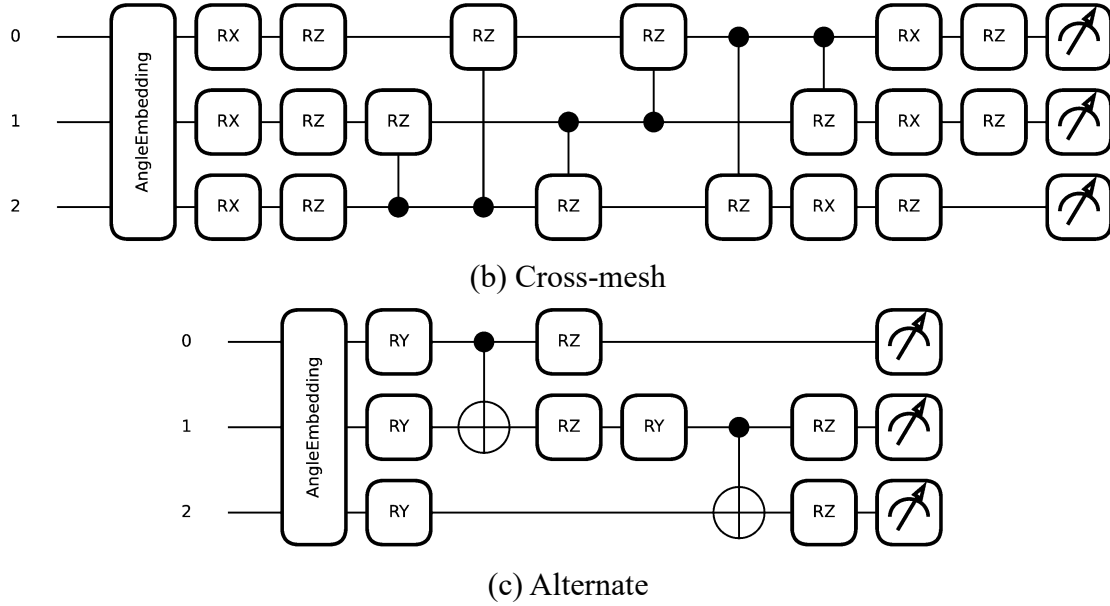


Fig. 2 Three Quantum Circuit Topologies used in this study.

3.2.3 Quantum Measurement

Quantum measurement is a key step in mapping quantum states to classical outputs. This paper adopts Pauli-Z expectation measurement: for each qubit i , the expected value $\langle Z_i \rangle$ under the Pauli-Z basis is measured, where Z is the Pauli-Z operator ($Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$). The expectation value is calculated as:

$$\langle Z_i \rangle = \langle \psi_{\text{final}} | Z_i \otimes I^{\otimes(n-1)} | \psi_{\text{final}} \rangle, \quad (18)$$

where $|\psi_{\text{final}}\rangle = U(\theta_2)|\psi_{\text{enc}}\rangle$ is the quantum state after passing through the PQC, and $U(\theta_2)$ is the total unitary transformation of the PQC, i.e., Eqs. (12), (14), or (17) depending on the topology.

The measurement results form a classical vector $\mathbf{Y}_q = [\langle Z_1 \rangle, \langle Z_2 \rangle, \dots, \langle Z_n \rangle]^T$, where each element ranges from $[-1, 1]$. This vector is then input to the classical postprocessor for physical variable mapping.

The quantum core can be abstractly expressed as:

$$\mathbf{Y}_q = f_2(\mathbf{X}_q, \theta_2), \quad (19)$$

where $f_2(\cdot)$ represents the mapping from the quantum input $\mathbf{X}_q$ and quantum parameters θ_2 to the quantum output $\mathbf{Y}_q \in \mathbb{R}^{d_q}$.

3.3 Classical Postprocessor

The classical postprocessor is responsible for quantum output decoding and physical variable mapping, converting the low-dimensional vector from quantum measurements into the target physical variables of reservoir seepage PDEs (pressure p , saturation S_w , or component concentration c). It is designed as a fully connected network with a single hidden layer, symmetric to the preprocessor: the input of the first layer (quantum output decoding) is the quantum measurement vector $\mathbf{Y}_q \in \mathbb{R}^{d_q}$, which is linearly transformed to the hidden layer vector $\mathbf{h}_3 \in \mathbb{R}^{d_h}$, i.e., $\mathbf{h}_3 = W_3 \mathbf{Y}_q + b_3$, where $W_3 \in \mathbb{R}^{d_h \times d_q}$ and $b_3 \in \mathbb{R}^{d_h}$ are trainable parameters initialized via Xavier. The Tanh activation function is introduced in this hidden layer, followed by a linear transformation through full connection to obtain the output layer data $\mathbf{Y}$. Thus, Eq. (9) summarizes the operation process of the classical postprocessor. For convenience, the set of parameters W_3 , W_4 , b_3 , b_4 involved in the classical postprocessor is denoted as θ_3 .

$$\mathbf{Y} = W_4 \text{Tanh}(W_3 \mathbf{Y}_q + b_3) + b_4 = f_3(\mathbf{Y}_q, \boldsymbol{\theta}_3), \quad (20)$$

where $W_4 \in \mathbb{R}^{1 \times d_h}$ and $b_3 \in \mathbb{R}^1$. f_3 represents the mapping from the output $\mathbf{Y}_q$ of the quantum network and the postprocessing network parameters $\boldsymbol{\theta}_3$ to the output $\mathbf{Y}$ of the postprocessing network.

Combining Eqs. (7), (8), and (9), the mapping of input $\mathbf{X} \in \mathbb{R}^{d_m}$ to output $\mathbf{Y} \in \mathbb{R}^1$ for the hybrid quantum-classical network in Fig. 1 can be denoted as:

$$\mathbf{Y} = f(\mathbf{X}, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3). \quad (21)$$

Physical loss is the core guarantee of QCPINN's physical consistency requiring simultaneous minimization of PDE residual loss (to ensure the solution satisfies the governing equation) and boundary/initial condition losses (to ensure the solution satisfies constraints). The total loss function is:

$$\mathcal{L}_{total}(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3) = \lambda_1 \mathcal{L}_{PDE}(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3) + \lambda_2 \mathcal{L}_{BC}(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3) + \lambda_3 \mathcal{L}_{IC}(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3), \quad (22)$$

where λ_1 , λ_2 , and λ_3 are loss weights (we set $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_3 = 1$ in this paper). The PDE residual loss is calculated using Mean Squared Error (MSE) for the residuals of the PDE governing equation based on automatic differentiation at sampling points. The boundary condition loss computes the MSE between predicted values and boundary conditions at boundary sampling points, covering both Dirichlet and Neumann boundaries. The initial condition loss calculates the MSE between predicted values and initial conditions at sampling points.

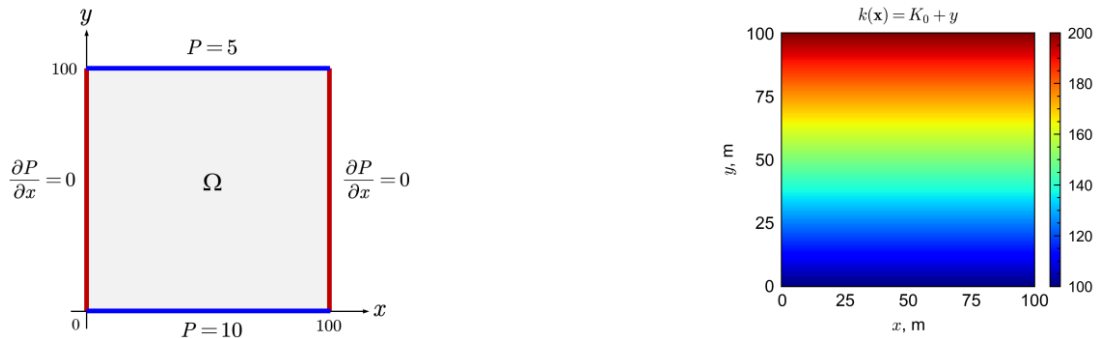
The parameters of the DV-circuit QCPINN include classical parameters $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_3$ as well as quantum parameters $\boldsymbol{\theta}_2$. The optimization process updates both types of parameters simultaneously. The Adam optimizer is adopted, combined with a ReduceLROnPlateau scheduler to dynamically decay the learning rate during loss plateaus. This reduces sensitivity to quantum parameter initialization and helps escape local optima. Additionally, gradient clipping is applied to limit the gradient norm, preventing gradient explosion of parameter gradients.

The DV-circuit QCPINN code in this paper is developed based on the open-source code by Farea et al. (2025) and the PennyLane platform. Key modifications and adaptations include reservoir PDE-specific adjustments and DV quantum circuit topology adaptations.

4 Numerical Experiments

4.1 Example 1

This example focuses on the heterogeneous-permeability single-phase flow problem described in Section 2.1. Consider a 2D rectangular reservoir $\Omega = [0, 100]m \times [0, 100]m$. By setting $k(\mathbf{x}) = K_0 + y$, $K_0 = 10^{-2}D$, $\mu = 1cp$, and $q_s(\mathbf{x}) = 0$ in Eq. (2). As shown in Fig. 3, the upper and lower boundaries of the reservoir in this example are constant-pressure boundaries, with the upper boundary pressure being 5 MPa and the lower boundary pressure 10 MPa. The left and right boundaries are set as impermeable closed boundaries.



(a) Reservoir domain and boundary conditions

(b) Heterogeneous permeability distribution

Fig. 3 Schematic of boundary conditions for the reservoir domain and heterogeneous permeability

distribution in Example 1.

The QCPINN model in this example takes spatial coordinates (x, y) as input and outputs the pressure value p . Table 1 lists the detailed parameters and training configurations of the QCPINN model (classical network and DV quantum circuit) used in this example. To systematically evaluate the impact of different quantum circuit topologies on QCPINN performance, this study adopted three quantum circuit topologies: Cascade, Cross-mesh, and Alternate, as shown in Fig. 4. During training, the Adam optimizer was used with an initial learning rate of 0.005. The ReduceLROnPlateau scheduler was introduced to reduce the learning rate when the loss showed no improvement for multiple consecutive epochs. The patience value was set to 1000 epochs, the learning rate decay factor was 0.9 (a 10% reduction each time), and the minimum learning rate was 10^{-6} (to prevent training stagnation due to excessively low learning rates). Meanwhile, the maximum parameter gradient norm was set to 1.0, and gradient clipping was applied to limit the gradient norm. To highlight the performance advantages of QCPINN, this example introduces a comparative analysis between the PINN solution and the QCPINN solution. To ensure fairness in the comparison, the PINN adopts a parameter configuration equivalent to that of QCPINN: the neural network architecture of the PINN is consistent with that of QCPINN in terms of the input and output layers, as well as the first and last hidden layers, each containing 30 neurons. The middle-hidden layer is set to 2 neurons, resulting in 273 trainable parameters for the PINN model, which is comparable to the 279 (Cascade), 283 (Cross-mesh), and 281 (Alternate) trainable parameters for three QCPINN models (including those of the preprocessing layer, quantum layer, and postprocessing layer). Meanwhile, the activation function, initial learning rate, batch size, optimizer type, and maximum number of training epochs of the PINN are exactly the same as those of QCPINN, eliminating the interference of training configuration differences on performance evaluation. It should be noted that all numerical experiments in this study were performed on a CPU platform with a system configuration of two Intel Xeon E5-2680 v4 processors (28 physical cores in total, 2.40 GHz main frequency) and 128 GB of memory. Although quantum simulation has inherent overhead on classical hardware, given the small scale of the quantum circuit in this example (2 qubits) and a batch size of 64, this CPU environment can still ensure stable and efficient training.

Fig. 5 presents a comparison of the loss function versus epochs during training for three quantum circuit topologies (Cascade, Cross-mesh, and Alternate) and the PINN. As shown in the figure, all three QCPINN topologies exhibit fast and stable convergence of the loss function, which drops rapidly from an initial value of approximately 10^2 and eventually stabilizes at the 10^{-3} order of magnitude, demonstrating the excellent trainability and loss function convergence of quantum topologies integrated with physical constraints. Among them, the Cross-mesh topology converges the fastest, with a significantly lower loss value in the later stage than the other two QCPINN topologies; the Cascade and Alternate topologies show a relatively gentle convergence rate but eventually reach a similar loss function level. In contrast, although the loss function of the PINN decreases rapidly in the early stage, its decline rate slows down in the later stage, and the final stabilized loss value is slightly higher than that of the other three QCPINN topologies, which highlights that the computational performance of the traditional PINN in this case is inferior to that of the QCPINN proposed in this study. Fig. 6 compares the reference solution, the PINN prediction results, and the QCPINN prediction results with the three quantum circuit topologies, and further shows the corresponding absolute error distributions. Visually, all models achieve pressure distribution solutions that are highly consistent with the reference solution. Compared with the QCPINNs with the three topologies, the deviation between the PINN and the reference solution is more significant, especially near the corner regions, indicating that the accuracy of the PINN is significantly lower than that of the QCPINNs. The quantitative error metrics are detailed in Table 2, from which we can see that all errors of the PINN are significantly higher than those of the QCPINN. For QCPINNs with different topologies, the Cascade topology has the lowest errors in all metrics and exhibits superior overall accuracy. Cross-mesh and Alternate show slight error clustering in the inner region and partial

boundaries, but still achieve higher computational accuracy than the PINN overall. In summary, for the heterogeneous reservoir seepage problem described by the elliptic equation in this example, although the QCPINNs with the three quantum circuit topologies exhibit slight differences in the loss function descent speed and computational accuracy, they all significantly outperform the traditional PINN. Among them, the Cascade topology performs better in terms of computational accuracy compared to the Cross-mesh and Alternate topologies. These results fully verify that the QCPINN integrated with quantum circuits demonstrates superior stability and accuracy advantages over the traditional PINN when dealing with problems involving single-phase flow in heterogeneous reservoirs.

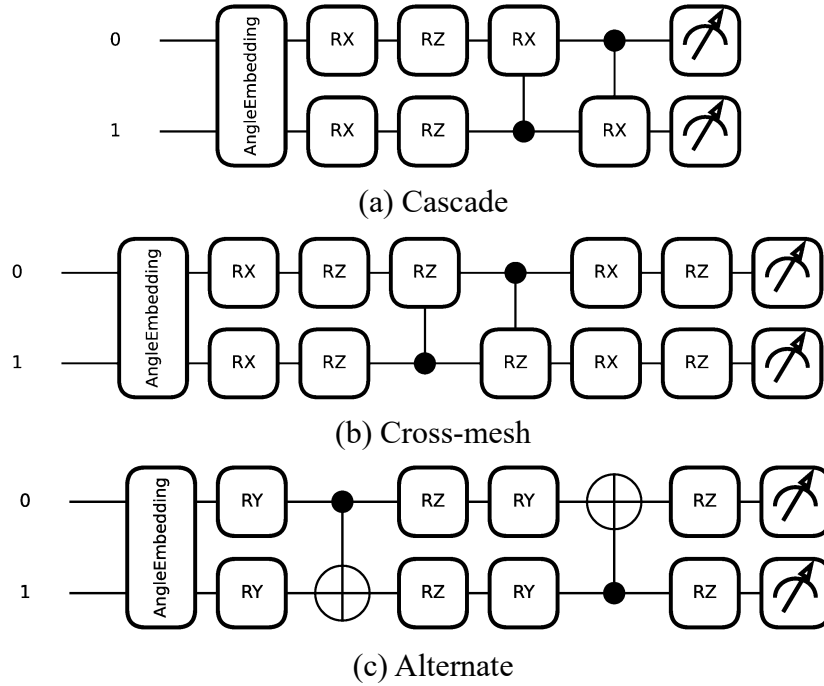


Fig. 4 Three quantum circuit topologies used in Example 1.

Table 1 Parameters of the neural networks and optimizers in Example 1.

Configuration Category	Parameter Name	Setting Value
Neural Network	Hidden Layers	1 layer with 50 neurons
	Activation Function	Tanh
DV Quantum circuit	Circuit Topology	Cascade
		Cross-mesh
		Alternate
	Number of Qubits	2
	Number of Quantum Layers	1
	Encoding	angle
Training Hyperparameters	Optimizer	Adam
	Initial Learning Rate	0.005
	Batch Size	64
	Maximum Training Epochs	20000

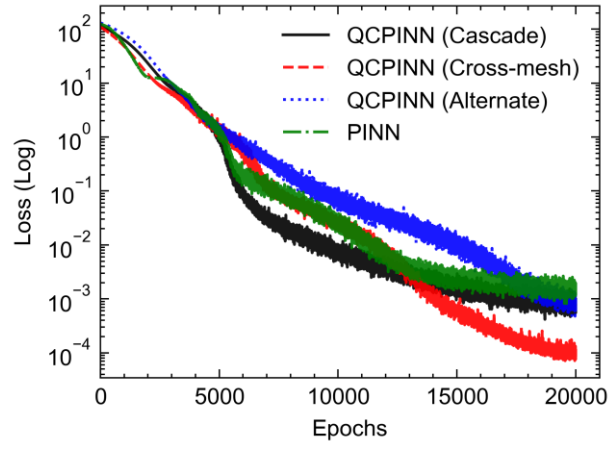


Fig. 5 Comparison of loss function versus epochs of QCPINN with three quantum circuit topologies and PINN in Example 1.

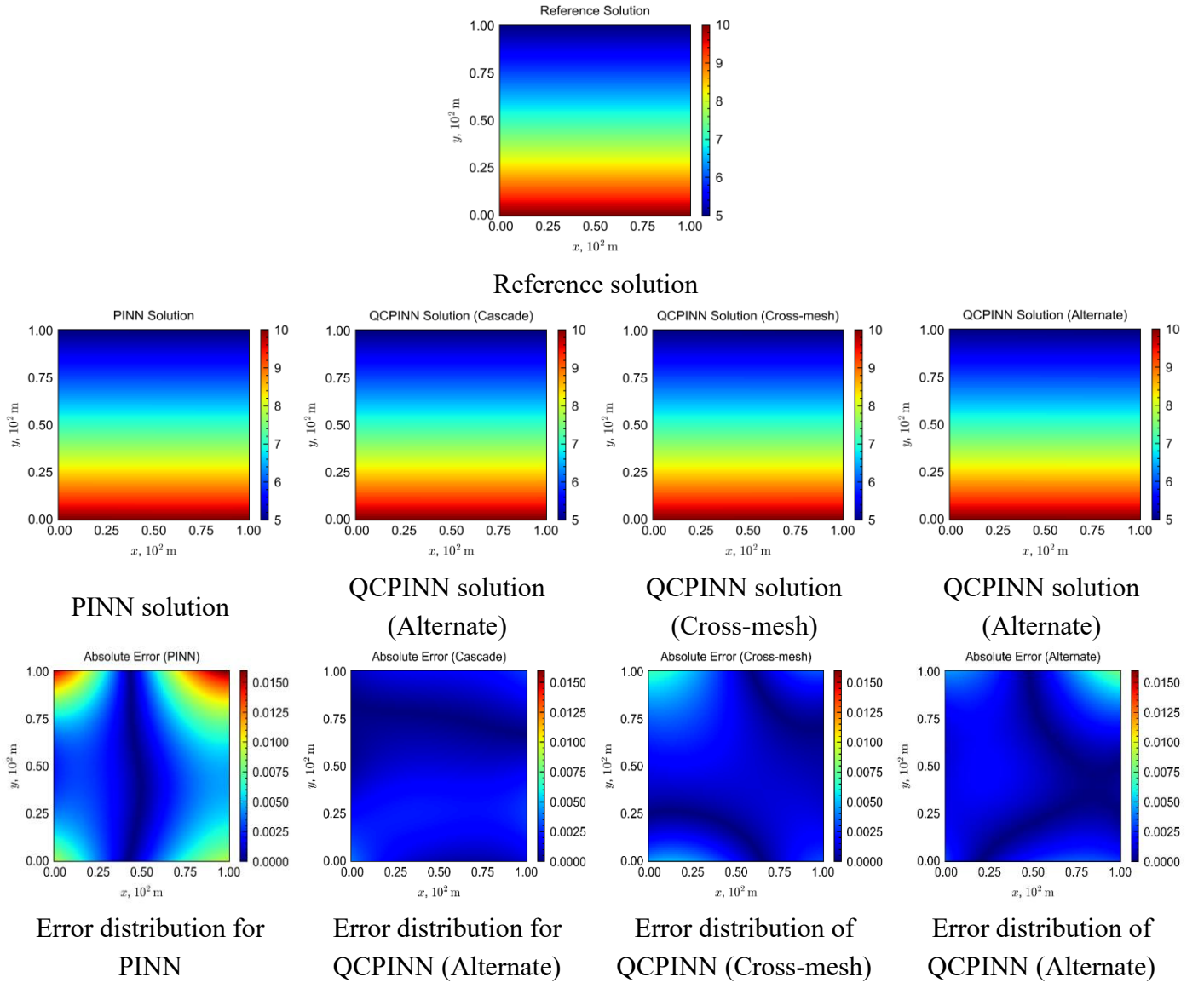


Fig. 6 Comparison of pressure distributions and absolute error distributions of QCPINN predictions using three topologies and PINN prediction in Example 1.

Table 2 Comparison of errors of the pressure profiles: PINN vs QCPINN using three quantum circuit topologies in Example 1.

Topological Structure	Mean Absolute Error	Max Absolute Error	Mean Relative Error	Max Relative Error
PINN	0.004125	0.016130	0.0629%	0.3226%
QCPINN (Cascade)	0.001295	0.003785	0.0174%	0.0579%
QCPINN (Cross-mesh)	0.001609	0.006753	0.0244%	0.1351%
QCPINN (Alternate)	0.001440	0.007903	0.0222%	0.1580%

4.2 Example 2

This example focuses on the two-phase flow problem described in Section 2.2. By setting $S_{wc} = 0$, $S_{or} = 0$, $n_o = n_w = 1$, and a constant total oil-water two-phase velocity $v = 1$ in the x -direction (the units of x , t , and v are adjusted to be compatible, so they are not explicitly specified in this example), Eq. (3) can be concretized as the nonlinear Buckley-Leverett (BL) equation in Eq. (23).

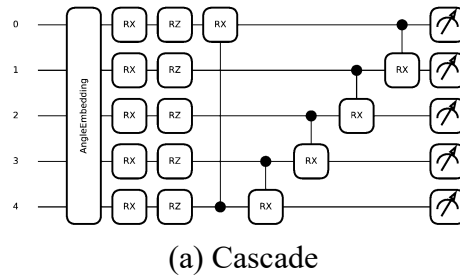
$$\frac{\partial S_w}{\partial t} + v \frac{\partial f_w}{\partial x} = 0, \quad 0 \leq x \leq 1, \quad 0 \leq t \leq 1, \quad f_w(S_w) = S_w / \left(S_w + \frac{1 - S_w}{M} \right) \quad (23)$$

where $M = 2$ is the viscosity ratio of oil to water.

The initial water saturation in the reservoir calculation domain is $S_{wc} = 0$. The left boundary is set as a constant water saturation boundary of 1 to simulate continuous water injection displacement conditions. The initial and boundary conditions of Eq. (23) are expressed as:

$$\begin{aligned} S_w(0, t) &= 1, \quad 0 \leq t \leq 1, \\ S_w(x, 0) &= 0, \quad 0 \leq x \leq 1. \end{aligned} \quad (24)$$

The QCPINN model in this example takes spatial coordinates and time variables (x, t) as input and outputs the water saturation S_w . To systematically evaluate the impact of different quantum circuit topologies on QCPINN performance, this example continues the design of Example 1 and uses the three topologies (Cascade, Cross-mesh, and Alternate) for calculations. Since this problem is a transient problem and the nonlinearity of the water cut function will theoretically increase the model learning difficulty, the number of qubits is increased to 5 while the number of quantum layers remains 1. The corresponding three quantum circuit topologies are shown in Fig. 7. The PINN model used for comparative validation adopts the same structure for its first and last hidden layers as QCPINN, each containing 50 neurons. To ensure a comparable number of trainable parameters between the two models, the middle hidden layer of the PINN is set to 5 neurons. As a result, the total number of trainable parameters in the PINN model is 756, while those in the three QCPINN models are 771, 796, and 772. In addition, key training parameters of the PINN, including input and output parameters, optimizer configuration, and the number of training iterations, are kept exactly the same as those of the QCPINN to ensure the fairness of the comparative experiments. All other configuration parameters of the two models (neural network structure, optimizer settings, etc.) and the hardware environment used for running are consistent with those in Example 1.



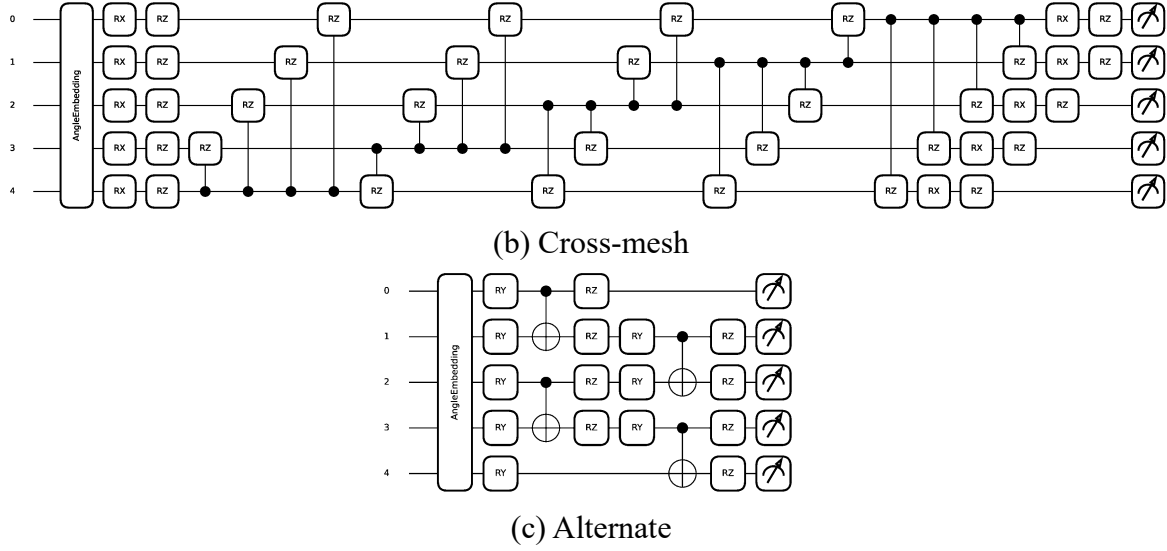


Fig. 7 Three quantum circuit topologies used in Example 2.

Fig. 8 presents a comparison of the loss function versus epochs during training for QCPINNs with three quantum topologies and the conventional PINN. Overall, all models achieve a gradual decrease in the loss function from an initial magnitude of 10^2 to 10^{-2} . However, the QCPINN models exhibit significantly smaller oscillations around 10^{-2} in the later training stages compared to the PINN. Notably, the loss function curve of the PINN displays intense and irregular fluctuations, reflecting its training instability when dealing with this strongly nonlinear transient problem. In contrast, although the three QCPINN models also exhibit some oscillations, their overall convergence trends are considerably more stable. Specifically, the QCPINN with the Cross-mesh topology shows the fastest loss decay in the early training stages; the QCPINN with the Cascade topology experiences relatively pronounced fluctuations in the later stages, posing certain challenges to the convergence of its loss function during training; the QCPINN with the Alternate topology demonstrates the smallest oscillation amplitude in its loss function curve, exhibiting the best training stability. Ultimately, the loss levels of the three QCPINN models converge to a consistent magnitude of 10^{-2} . While the final loss of the PINN also approaches this magnitude, its pervasive fluctuating behavior throughout training indicates inferior convergence compared to the QCPINNs.

Fig. 9 compares the prediction results of the three QCPINN topologies at different time instants ($t=0.1, 0.3, 0.5, 0.7, 0.9$) with the reference solution from the upwind finite difference method based on an ultra-dense grid. The results demonstrate that all three topologies can accurately capture the steep water-flooding front and its migration behavior, with the overall morphology being basically consistent with the reference solution. Specifically, the Cascade topology shows slight deviations in the front and the water-unswept regions, but the overall curve trend remains consistent with the reference solution. The Cross-mesh topology is highly consistent with the reference solution in all regions at each time step except for a noticeable deviation from the reference solution at the front. The Alternate topology only exhibits minor deviations in the water-unswept regions, while fitting most closely with the reference solution in other regions, almost overlapping completely.

Fig. 10 presents the relative L2 errors between the saturation predictions of QCPINN using three quantum-circuit topologies, PINN and the reference solution over the time interval of 0.1-0.9. As a quantitative evaluation metric, the L2 error is defined as follows:

$$L_2(t_k) = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_{pred}(x_i, t_k) - S_{ref}(x_i, t_k))^2}, \quad (25)$$

where N denotes the number of spatial sampling points, t_k represents a specific time point, S_{pred} and S_{ref} stand for the water saturation predicted by QCPINN, PINN and the reference solution, respectively.

We can see from Fig. 10 that the L2 error of the PINN remains consistently higher than those of the QCPINN models using three topologies across all time steps except $t=0.1$, with a maximum error approaching 0.0225. The errors of the three QCPINN models remain at a low magnitude, indicating the QCPINN exhibits excellent accuracy. Specifically, in the early stage $t=0.1$, the Cross-mesh topology achieves the lowest error (approximately 0.008), followed by the Cascade topology, while the Alternate topology yields the highest error. With the time increasing, the accuracy performance of each topology undergoes significant changes. The error of the Cross-mesh topology gradually increases, whereas the error of the Alternate topology decreases markedly and remains consistently at a low level (around 0.006) for all time steps from $t=0.4$ to $t=0.9$, demonstrating significantly superior accuracy compared to the Cascade and Cross-mesh topologies. The Cascade topology exhibits relatively high errors across all time steps and fails to demonstrate obvious advantages at any stage.

Overall, for the 1D nonlinear Buckley-Leverett (BL) equation describing waterflooding, compared with PINN, the QCPINN models constructed with the Cascade, Cross-mesh, and Alternate topologies all demonstrate better capabilities in capturing physical characteristics and higher prediction accuracy. Among them, the Alternate topology stands out in terms of training convergence and computational accuracy, showing relatively superior performance.

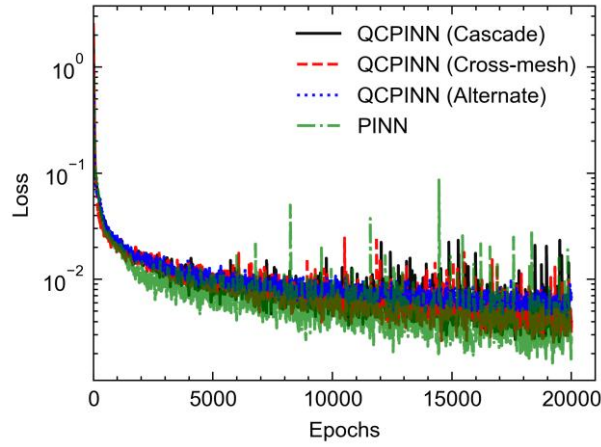
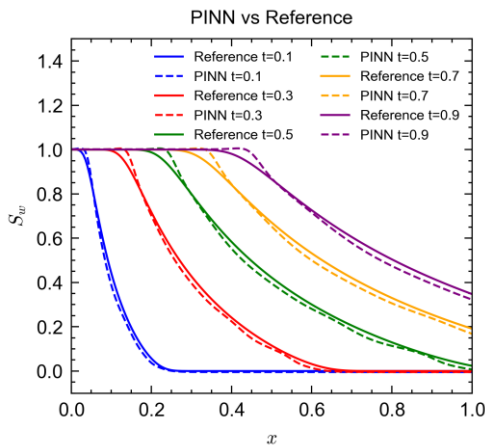
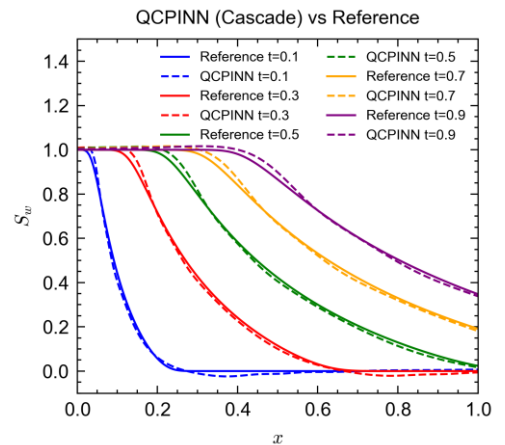


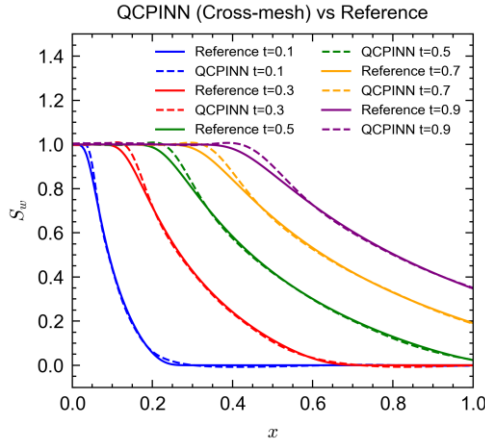
Fig. 8 Comparison of loss function versus epochs of QCPINN with three quantum circuit topologies and PINN in Example 2.



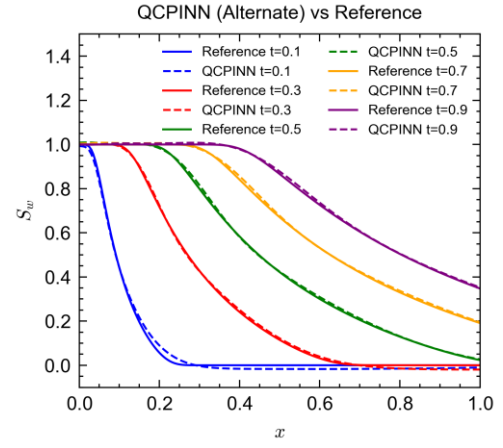
(a) PINN solution



(b) QCPINN solution (Cascade)



(c) QCPINN solution (Cross-mesh)



(d) QCPINN solution (Alternate)

Fig. 9 Comparison of saturation distributions: QCPINN predictions using three topologies, PINN vs reference solution at different times.

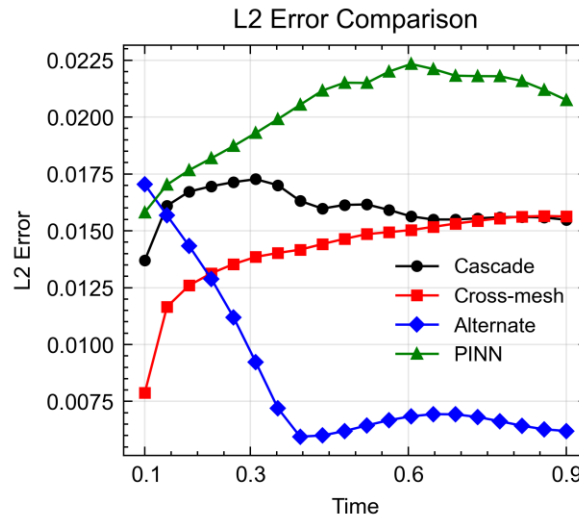


Fig. 10 Relative L2 error comparison of three topologies in Example 2.

4.3 Example 3

This example focuses on the component transport problem considering adsorption as discussed in Section 2.3. As illustrated in Fig. 11, a rectangular reservoir computational domain $\Omega = [0, 100]\text{m} \times [0, 100]\text{m}$ is still adopted. In Eq. (6), suppose $\phi = 0.3$, $\rho_b = 2000 \text{ kg/m}^3$, $K_d = 1.0 \times 10^{-4} \text{ m}^3/\text{kg}$, $D = 0.5 \text{ m}^2/\text{s}$, and $\mathbf{v} = (0.5, 0) \text{ m/s}$. At the initial moment, the component concentration throughout the reservoir is 0 kg/m^3 . A Dirichlet boundary condition is applied to the left boundary, where the solute concentration is maintained at 1 kg/m^3 ; the right boundary has a zero diffusion flux but no restriction on advective flux. The upper and lower boundaries are set as no-flux closed boundaries. The physical process in this example describes the migration behavior of components gradually propagating along the main flow direction under the combined effects of convection, dispersion, and linear adsorption. Due to the retardation effect of linear adsorption, the effective migration velocity of components is lower than the seepage velocity. In the initial stage, components migrate from the high-concentration region of the left boundary to the low-concentration region, forming an adsorption-retarded concentration front that advances along the x -direction over time. Meanwhile, dispersion causes the front to gradually spread during propagation, overall reflecting the mass transport characteristics of convection dominance, adsorption retardation, and dispersion broadening, which fully couples the multi-physical processes of advective transport, molecular-mechanical dispersion, and solid-phase linear adsorption.

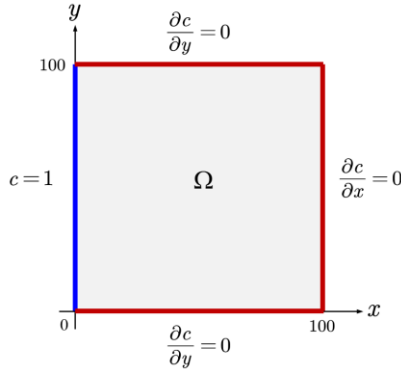


Fig. 11 Schematic of the computational domain and boundary conditions for Example 3.

The QCPINN model takes spatiotemporal coordinates (x, y, t) as input and outputs the target concentration c . Since this example involves multi-physical processes, to accurately characterize the component migration behavior, 6 qubits are used in this case, and three quantum topologies (Cascade, Cross-mesh, and Alternate) are still adopted as shown in Fig. 12. The remaining parameters and training configurations of the model are the same as those in Example 2. To demonstrate the performance advantages of QCPINN, a PINN model is also introduced as a comparator in this example. The input layer, output layer, and first/last hidden layers of this PINN are structurally identical to those of QCPINN. By setting the number of neurons in the middle-hidden layer to 6, the total number of trainable parameters in the PINN reaches 907, which is comparable to the 925, 961, and 931 trainable parameters of QCPINN, ensuring fairness in the comparison. In addition, key training parameters of the PINN, such as optimizer configuration and the number of training iterations, are kept consistent with those of QCPINN to eliminate the influence of differences in training conditions on performance evaluation.

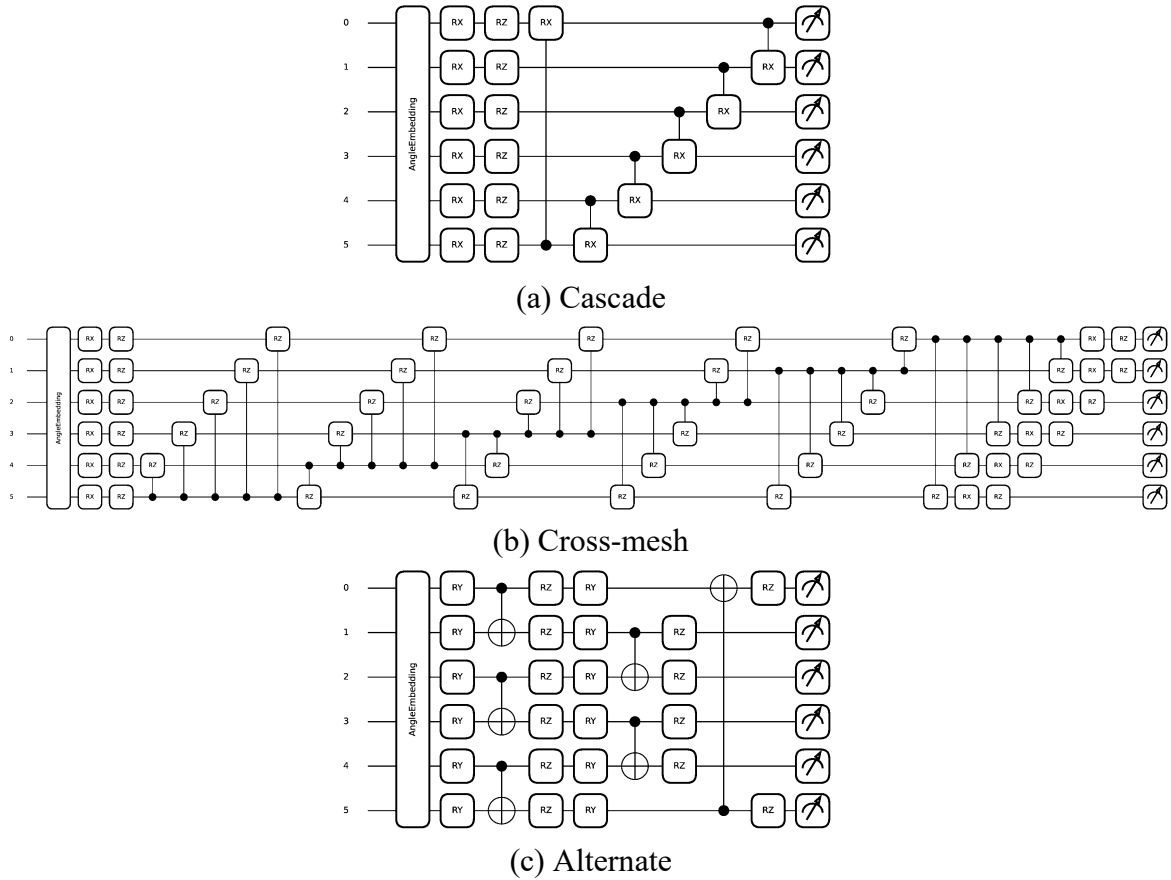


Fig. 12 Three quantum circuit topologies used in Example 3.

Fig. 13 shows the convergence process of the loss function during training for QCPINN using the three quantum circuit topologies (Cascade, Cross-mesh, and Alternate) and the traditional PINN. Overall, all models exhibit good convergence performance: the loss value decreases rapidly with the increase in the number of iterations, dropping from the initial order of 10^0 to around the order of 10^{-2} . The loss function curve of the PINN exhibits relatively severe fluctuations and shows no trend of steady convergence even in the later stages of training. In contrast, the loss function curves of the three QCPINN models are much smoother, with significantly smaller oscillation amplitudes than the PINN, demonstrating superior training stability. Among three QCPINN models, the Cascade and Cross-mesh topologies show consistent descending trends, with faster convergence speeds and slightly lower final loss function values compared to the Alternate topology. In contrast, the Alternate topology has a relatively higher loss function value in the later stage of training.

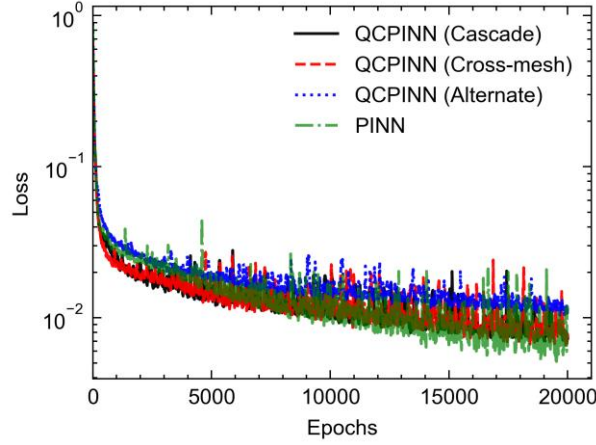


Fig. 13 Comparison of loss function versus epochs of QCPINN with three quantum circuit topologies and PINN in Example 3.

Fig. 14 shows the comparisons of concentration distributions between the reference solution and the predictions of the QCPINNs and the PINN at $t=0.8$, and further presents the corresponding absolute error distributions. It can be seen that the main errors of all models are concentrated in the component displacement front, where the concentration varies sharply. This is because the large concentration gradient in this region significantly increases the approximation difficulty for neural networks. A detailed comparison shows that the absolute error of the PINN in the front region is markedly higher than that of the three QCPINN models, with its peak error approaching 0.12, reflecting its insufficient accuracy in capturing regions with high physical gradients. In contrast, the three QCPINN models exhibit overall lower error levels. Among them, the QCPINN with the Cascade topology shows the most uniform error distribution and the smallest peak error (approximately 0.06) in the component displacement front, demonstrating the best local approximation capability.

Fig. 15 further compares the evolution of the L2 error for the four models over the time interval from $t=0.3$ to $t=0.6$. This stage corresponds to the period when the component displacement front is located in the middle of the computational domain, during which the computed L2 error best reflects the differences in numerical errors among different models. Otherwise, regions where the predictions of different models show little difference would dominate the L2 error, masking the discrepancy between them. As observed from Fig. 15, the L2 error of the PINN is consistently and significantly higher than those of the three QCPINN models throughout the time interval, while the L2 errors among the three QCPINN models are relatively close. As mentioned earlier, since the predictions of different models exhibit almost no difference in regions outside the component displacement front, the discrepancy in L2 error is mainly contributed by the displacement front, which accounts for a small portion of the entire computational domain. Therefore, although the absolute value of the additional error of the PINN relative to the three QCPINN models is not large, it clearly indicates that the prediction accuracy of QCPINNs is substantially higher than that of the PINN. This also suggests that the

advantage of QCPINNs over the PINN is inherent to quantum neural networks compared with conventional neural networks, and is independent of the specific quantum circuit topology adopted. Furthermore, among the three different quantum circuit topologies, the L2 errors of the Cross-mesh and Alternate topologies are similar. The L2 error of the Cascade topology is slightly higher than the other two topologies when $t \leq 0.44$, but becomes noticeably lower when $t > 0.44$. Overall, the Cascade topology achieves the lowest L2 error level and the highest computational accuracy.

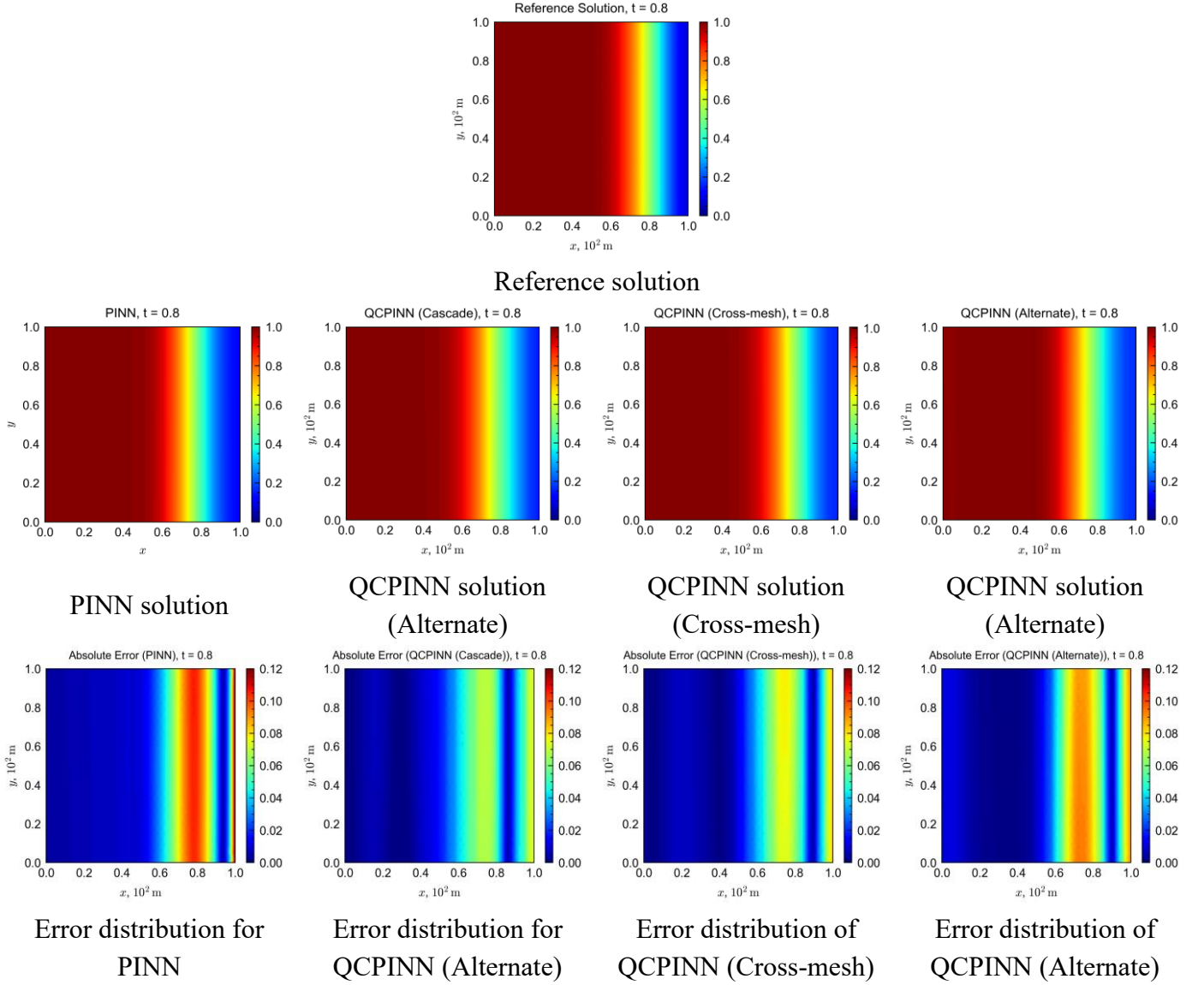


Fig. 14 Comparison of concentration distributions and absolute error distributions of QCPINN predictions using three topologies, PINN prediction vs reference solution at $t=0.8$ in Example 3.

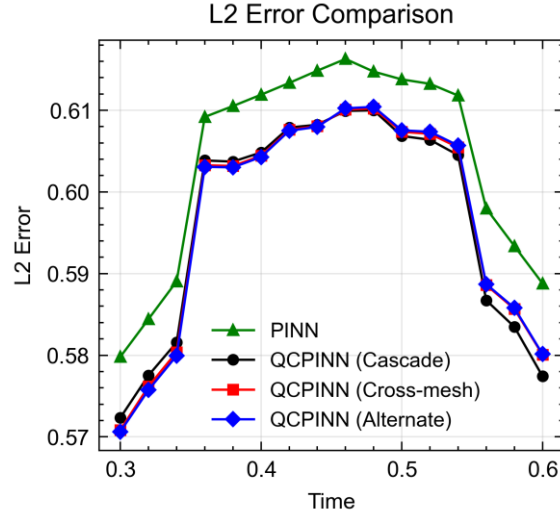


Fig. 15 Relative L2 error comparison: QCPINN with three topologies vs PINN in Example 3.

4.4 Example 4

This example focuses on the two-phase oil-water seepage problem in heterogeneous reservoirs where pressure and water saturation need to be calculated synchronously, which is different from solving the Buckley-Leverett equation that only calculates water saturation with compressibility neglected in Example 2. Compared with Example 2, two equations in Eq. (26) need to be handled synchronously in this example, which are the mass conservation equations for the water phase and oil phase, respectively. The independent variables are x , y , and t , and the dependent variables include both pressure and water saturation. In addition, the exponential strongly heterogeneous permeability distribution in Eq. (27) is introduced, which represents the most common and classical two-phase flow scenario in reservoir engineering. The only difference is that the singular source-sink term caused by injection-production wells, which is common in reservoir engineering, is not added in this example. This is because numerous previous studies (Karniadakis et al., 2021; Zhang et al., 2022; Han et al., 2023; Rao et al., 2025b) have shown that conventional PINN methods that only rely on automatic differentiation to construct equation residual terms without additional processing cannot effectively handle singular source-sink terms. At present, some improved PINNs are generally adopted to deal with such singular source-sink terms. For instance, the PINN based on domain decomposition (PINN-DD) that divides the reservoir domain into well-containing and well-free sub-domains (Han et al., 2023; Kharazmi et al., 2021), or PINNs based on numerical method formats with integral processing, such as the boundary-integral neural network (BINN) based on the boundary element method (Rao et al., 2025c), the variational physics-informed neural network (VPINN) based on the finite element method (Kharazmi et al., 2019), the enriched-PINN (E-PINN) based on the finite volume method (Yan et al., 2024), or some regularization methods that approximate the singular Dirac function with Gaussian functions or truncated polynomial functions. Among the above processing methods, only the regularization method is directly compatible with the conventional PINN framework, but its computational performance is generally poor. Therefore, the regularization method is not used to add the singular source-sink term caused by injection-production wells in this example, because the large error inherent in the regularization method will significantly weaken the illustration of the computational performance advantages of the QCPINN proposed in this work. Other processing methods for singular source-sink terms require underlying modifications to the conventional PINN framework, and such modifications will completely alter the method architecture in Section 3 of this paper. Thus, adding domain decomposition, boundary integral, weak form or finite volume processing for singular source-sink terms on the basis of this work to form the corresponding BIQCPINN, QCPINN-DD, VQCPINN or E-QCPINN is fully worthy of a complete new research. Hence, this example focuses on the classical two-phase oil-water flow simulation in heterogeneous reservoirs without source-sink terms to fully illustrate the performance advantages of the

quantum computing-coupled QCPINN over the traditional PINN. It can be fully expected that since PINN-based methods such as BINN, PINN-DD, VPINN and E-PINN can achieve excellent computational performance for seepage problems involving singular source-sink terms, BIQCPINN, QCPINN-DD, VQCPINN, and E-QCPINN which are based on the more superior QCPINN, will also effectively handle seepage problems with singular source-sink terms.

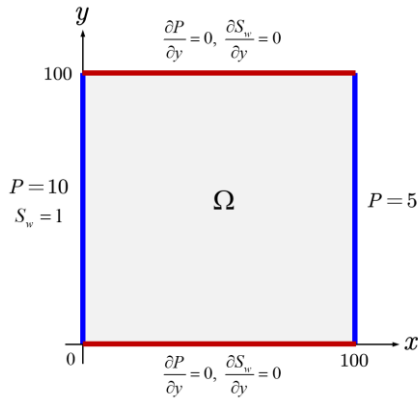
$$\phi \frac{\partial S_w}{\partial t} + \nabla \cdot \left(\frac{k(x, y) k_{rw}(S_w)}{\mu_w} \nabla p \right) = 0, \quad (26)$$

$$\phi \frac{\partial S_o}{\partial t} + \nabla \cdot \left(\frac{k(x, y) k_{ro}(S_w)}{\mu_o} \nabla p \right) = 0,$$

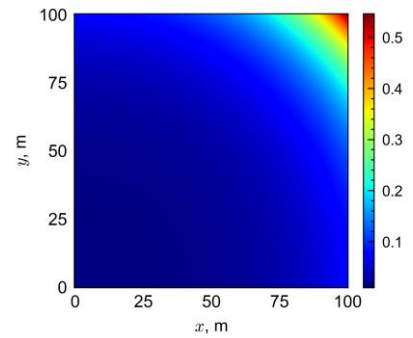
$$k(x, y) = 0.01 \times e^{\frac{x^2 + y^2}{5000}} D, \quad (27)$$

where S_w is the water saturation and p is the oil phase pressure; $k(x, y)$ is the absolute permeability, which is a function of spatial coordinates. $k_{rw}(S_w)$ and $k_{ro}(S_w)$ are the relative permeabilities of the water phase and oil phase, respectively, which are nonlinear functions of saturation and described by the Corey model in Eq. (5). For the convenience of comparative analysis, $n_w = 1$, $n_o = 1$, $\mu_w = 1cp$, and $\mu_o = 1cp$ are taken in this example.

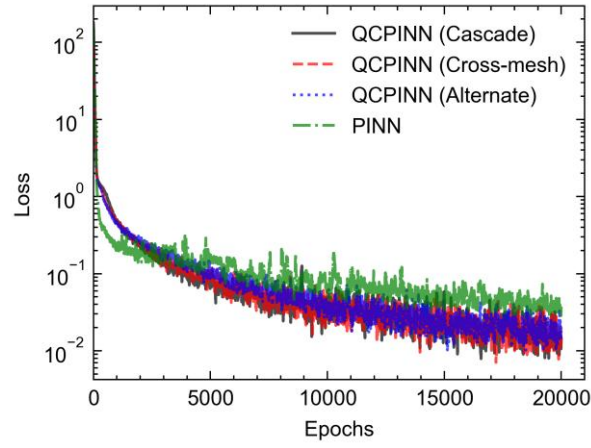
As shown in Fig. 16(a), the computational domain of the two-dimensional reservoir in this example is $\Omega = [0, 100]m \times [0, 100]m$. For the initial conditions, $S_w(x, 0) = S_{wc}$ and $S_{wc} = 0$ are adopted in this example. The upper and lower boundaries are closed boundaries satisfying the no-flow condition with $\nabla S_w(x, t) \cdot \mathbf{n} = 0$ and $\nabla p(x, t) \cdot \mathbf{n} = 0$. The left boundary is a constant-pressure water injection boundary with a pressure of 10 MPa and $S_w(0, t) = 1 - S_{or}$, where $S_{or} = 0$ are taken in this example, and the pressure of the right boundary is 5 MPa. As shown in Fig. 16(b), the exponential permeability distribution in Eq. (2) is adopted in this example, which generates a drastic change of more than an order of magnitude in the computational domain to fully verify the computational performance of QCPINN.



(a) Reservoir domain and boundary conditions



(b) Heterogeneous permeability distribution



(c) Training loss over epochs

Fig. 16 Schematic of boundary conditions for the reservoir domain, heterogeneous permeability distribution and training loss over epochs of Example 4.

Due to the significant differences in physical magnitudes and dynamic characteristics between pressure and saturation, to avoid the gradient imbalance problem caused by inconsistent output scales of a single network, two independent QCPINN sub-networks are constructed in this example to approximate pressure and saturation, respectively. The configuration of the quantum core of these networks (including the number of qubits, layers and quantum circuit topology) is the same as that in Example 3. The losses of the PDE residual, boundary conditions and initial conditions for Eq. (26) constituted by the two sub-networks are calculated separately and then summed, and the optimization is performed by synchronously updating the parameters of the two networks. The parameter settings such as optimizer, learning rate strategy and batch size are consistent with those in Example 3.

To quantitatively evaluate the performance improvement of QCPINN over PINN in this example, a classical PINN model for comparison is trained simultaneously, as in the previous examples. The structure of the input layer, output layer and the first and last hidden layers of the PINN model is exactly the same as that of QCPINN, and the number of neurons in the middle hidden layers is 6, the same as the number of qubits in QCPINN, to ensure a comparable scale of trainable parameters. The PINN also uses two independent sub-networks to output pressure and saturation, respectively, to ensure the fairness of comparison.

Fig. 16(c) shows the loss functions versus epochs of the QCPINN models with three quantum circuit topologies and the traditional PINN model during the training process. Overall, the loss values of all models gradually decrease from the initial order of 100 to about the order of 10^{-2} . However, the loss function value of the traditional PINN remains at a relatively higher level than those of the QCPINNs after 3000 epochs, and its loss function convergence ability is significantly weaker than that of the three QCPINN models. The convergence of the three QCPINN models is relatively consistent with similar final loss function values, indicating that QCPINN indeed achieves better training effects than PINN when dealing with the more complex heterogeneous two-phase oil-water seepage process in the physical space in this example.

Fig. 17 compares the predicted oil phase pressure distributions and the absolute error distributions of the pressure predictions by the three QCPINNs and the traditional PINN with the reference solution at $t=0.82$ (according to unit conversion, one unit time in this example corresponds to $100000/864 \approx 115.7407$ days). From the perspective of absolute error distribution, the absolute error of the pressure prediction by PINN is significantly higher overall than that of the three QCPINN models, with its error peak exceeding 0.117 MPa, while the absolute errors of the pressure predictions by the three QCPINN models are all less than 0.039 MPa.

Fig. 18 compares the predicted water saturation distributions and the absolute error distributions of the saturation predictions by the three QCPINNs and PINN with the reference solution at $t=0.82$. In terms of saturation distribution, the results of the QCPINN models are obviously more consistent with the reference

solution than those of the PINN model, and the numerical diffusion of the QCPINN results at the water flooding front is significantly smaller than that of PINN. Among them, the discontinuity degree of the water flooding front predicted by the QCPINN with Alternate topology is even better than that of the high-precision numerical solution calculated with a $200 \times 200 = 40000$ grids. The error distribution diagrams also clearly reflect that the absolute error values of PINN are significantly higher, which demonstrates the superior performance of QCPINN.

Fig. 19 shows the evolution law of the L2 error of the oil phase pressure distribution and water saturation distribution predicted by the three QCPINN models and PINN within the time range of 0-1. It can be seen that the error of PINN remains at the highest level throughout the time range of 0-1, while the L2 errors of the three QCPINN models are significantly lower than that of PINN all the time. Among them, the average error level of the Cross-mesh topology is the lowest among the three quantum circuit topologies. In terms of pressure distribution, the L2 error of PINN is nearly 20 times that of QCPINN at maximum; in terms of saturation distribution, the L2 error of PINN is nearly 5 times that of QCPINN at maximum.

In general, for the two-phase flow problem with strongly heterogeneous permeability distribution and full pressure-saturation coupling in this example, the three QCPINN models all exhibit better training convergence and prediction accuracy compared with PINN, which fully demonstrates the important significance of the QCPINN studied in this paper to replace PINN and introduce quantum computing into the field of classical models and reservoir engineering.

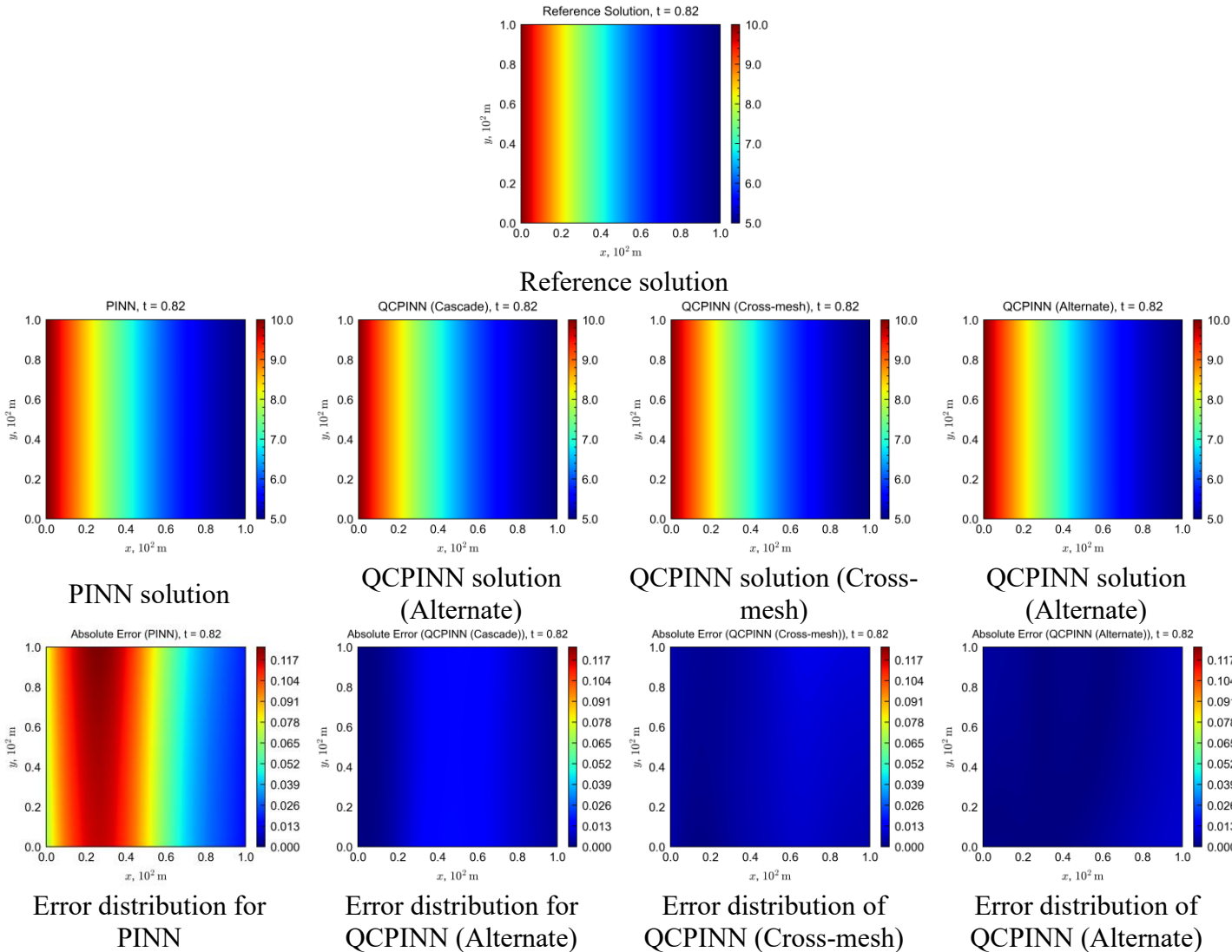


Fig. 17 Comparison of pressure distributions and absolute error distributions of QCPINN predictions using three topologies and PINN prediction at $t=0.82$ in Example 4.

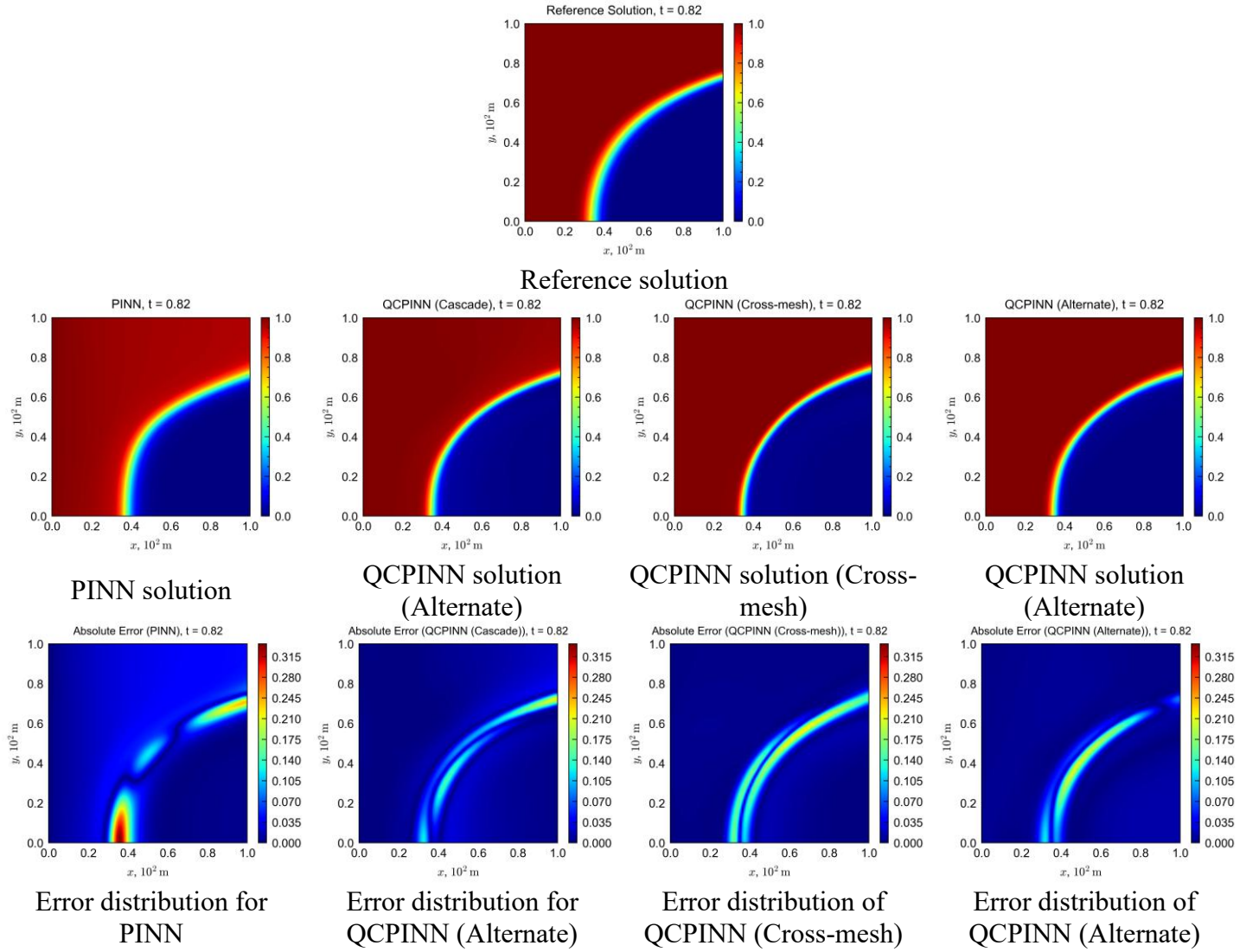


Fig. 18 Comparison of water saturation distributions and absolute error distributions of QCPINN predictions using three topologies and PINN prediction at $t=0.82$ in Example 4.

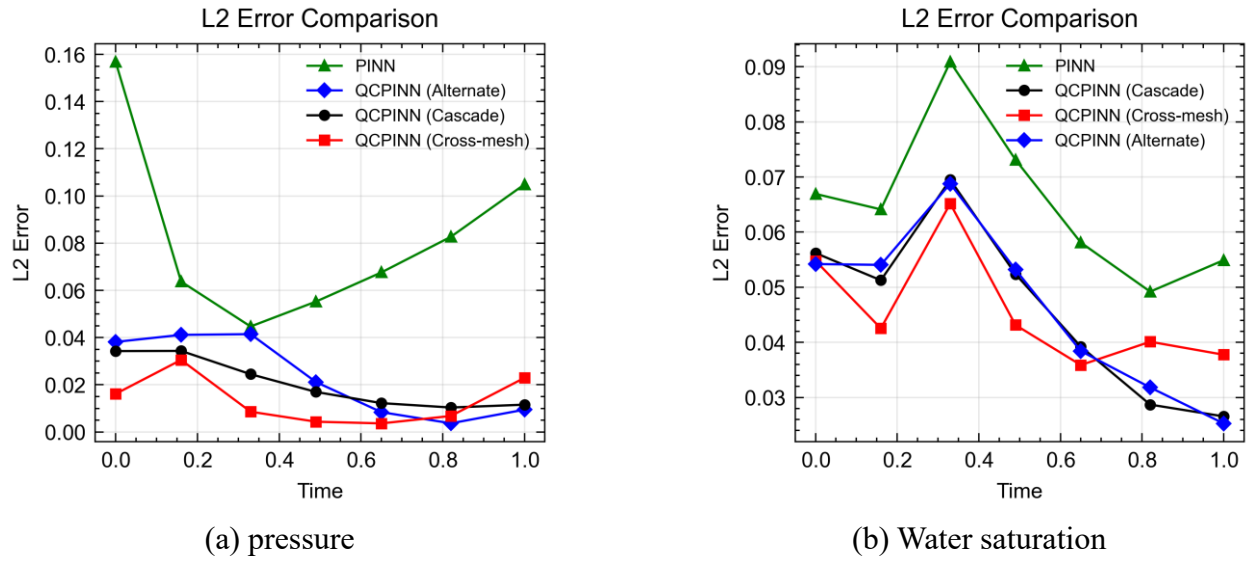


Fig. 19 Comparison of L2 errors of pressure and water saturation predictions: QCPINN with three topologies vs PINN in Example 4.

4.5 Discussion on the Relationship Between Quantum Circuit Topology and Physical Problem Characteristics

The observed differences in optimal quantum circuit topologies across the four seepage PDEs invite a discussion on the potential relationship between circuit design choices and the inherent mathematical and physical nature of the governing equations. While our study is exploratory and not intended to establish definitive causal rules, we offer the following interpretative observations that may guide future research in quantum-classical co-design for reservoir simulation.

The elliptic pressure diffusion equation in Eq. (2) describes a spatially smooth field with localized heterogeneities, where pressure distribution is dominated by permeability spatial variation and boundary constraint propagation. The Cascade topology, with its ring-structured entanglement and sequential single-qubit rotation-entanglement layers, demonstrates superior accuracy in this scenario. A plausible interpretation is that the cyclic entanglement pattern of the Cascade topology effectively encodes the spatial correlation of permeability heterogeneities, aligning with the local support of Darcy's law where pressure gradients depend on adjacent permeability values. In contrast, the Cross-mesh topology's full-connectivity entanglement may introduce redundant long-range couplings that do not correspond to the equation's local physical mechanism, leading to slight error clustering. The Alternate topology's nearest-neighbor entanglement, while stable, exhibits lower parameter efficiency compared to the Cascade topology in capturing the continuous pressure diffusion process.

The hyperbolic Buckley-Leverett (BL) equation in Eq. (3) is characterized by a traveling shock front, requiring the network to capture sharp, transient discontinuities and nonlinear saturation-conductivity relationships. The Alternate topology, with its alternating local rotation and CNOT entanglement sequence, achieves the most balanced performance in convergence stability and front-capturing accuracy. This advantage may stem from its ability to represent piecewise smooth functions through localized entanglement, avoiding the front-smearing effect caused by the Cascade topology's cyclic dependencies. The Cross-mesh topology, despite its high theoretical expressivity, shows increasing errors in later transient stages, possibly due to overfitting the dynamic shock front evolution or inefficient parameter optimization induced by full-connectivity entanglement. These observations suggest that staggered local entanglement patterns are well-suited for hyperbolic equations with strong nonlinearity and moving discontinuities.

The convection-diffusion-adsorption equation in Eq. (6) couples three core physical processes: global advective transport, local dispersive smoothing, and adsorption-induced retardation (i.e., a memory effect). The Cascade topology's ring entanglement structure outperforms other topologies in this multi-physics coupling scenario. This superiority may be attributed to its ability to encode both local dispersion effects and global advective correlations along the flow direction, effectively mimicking the combined action of mass transport and adsorption retardation. The Alternate topology's local entanglement constraint limits its ability to capture long-range advective coupling, leading to slightly higher errors in the concentration front region. The Cross-mesh topology, while capable of modeling complex multi-physics correlations, exhibits noisier convergence due to its higher parameter count and increased sensitivity to gradient optimization in classical simulation environments.

The fully coupled pressure-saturation two-phase oil-water seepage equation for heterogeneous reservoirs in Eq. (26) represents the most complex scenario in this study, integrating elliptic pressure diffusion, hyperbolic saturation advection, exponential permeability heterogeneity, and strong cross-coupling between pressure and saturation fields. The Alternate topology emerges as the optimal choice for this problem, delivering the sharpest capture of water flooding fronts (even outperforming high-precision numerical solutions with ultra-dense grids) and the lowest saturation prediction errors. Its staggered local entanglement design uniquely balances the need to resolve discontinuous saturation fronts (a hyperbolic characteristic) and smooth pressure field variations (an elliptic characteristic) in a single coupled framework, avoiding the gradient imbalance and numerical diffusion that plague classical PINNs and other quantum topologies. The Cross-mesh topology exhibits the most balanced L2 error performance for both pressure and saturation in this

coupled problem, with its full-connectivity entanglement effectively encoding the cross-field correlations between pressure and saturation without significant overfitting. The Cascade topology maintains competitive accuracy for pressure distribution but shows slight smearing in saturation front capture, as its cyclic entanglement is better tailored for single-field spatial correlation than multi-field cross-coupling.

Our experiments suggest a tentative, problem-dependent trade-off between entanglement structure and physical process adaptability. For elliptic equations dominated by spatial heterogeneity and local physical constraints (e.g., single-phase pressure diffusion), ring-structured entanglement (Cascade topology) is preferred for its balanced expressivity and parameter efficiency. For hyperbolic equations with sharp transient discontinuities (e.g., two-phase BL equation), staggered local entanglement (Alternate topology) excels in capturing nonlinear front dynamics while avoiding numerical diffusion. For multi-physics coupled equations involving both local and global processes (e.g., convection-diffusion-adsorption), cyclic entanglement (Cascade topology) effectively integrates diverse physical mechanisms, outperforming other topologies in overall accuracy. For fully coupled multi-field seepage problems with concurrent elliptic-hyperbolic characteristics and strong heterogeneity (e.g., heterogeneous coupled pressure-saturation two-phase flow), the Alternate topology is optimal for front capture and saturation prediction, while the Cross-mesh topology provides balanced performance across all coupled physical fields. The full-connectivity Cross-mesh topology, despite its high theoretical expressivity, does not consistently yield superior results across all scenarios, likely due to NISQ-era constraints such as limited quantum resources and classical simulation overhead, but its balanced performance makes it a robust choice for coupled multi-physics problems.

These observations are preliminary and warrant further systematic investigation. Future work could scale the number of qubits and circuit depth, test more diverse PDE classes, and explore hardware-aware noise resilience. Nevertheless, these findings underscore the promise of tailoring quantum circuit topologies to the physical character of the target PDE. This direction could enrich the co-design of quantum-classical algorithms for industrial-scale reservoir simulation.

5 Conclusions

This study pioneers the adaptation and application of Quantum-Classical Physics-Informed Neural Networks (QCPINNs) to reservoir numerical simulation, systematically addressing four core seepage PDEs with heterogeneity, strong nonlinearity, convection dominance and full pressure-saturation coupling in reservoir engineering. Key conclusions are drawn as follows:

The adapted DV-Circuit QCPINN framework effectively integrates quantum computing advantages with physical constraints. By combining classical networks for feature preprocessing/decoding and a quantum core for high-dimensional mapping, it overcomes the parameter inefficiency, nonlinear fitting limitations and gradient imbalance problems of classical PINNs, achieving accurate PDE solutions with reduced computational complexity. Different quantum circuit topologies exhibit distinct and problem-adaptive performance in various seepage scenarios with clear optimal application scopes: The Alternate topology is the optimal choice for heterogeneous single-phase flow, transient nonlinear two-phase BL equation, and heterogeneous fully coupled pressure-saturation two-phase flow, excelling in capturing sharp discontinuous fronts, resolving strong permeability heterogeneity, and balancing elliptic-hyperbolic characteristics; The Cascade topology excels in multi-physics coupled compositional flow with convection-dispersion-adsorption, effectively encoding the combination of local dispersion and global advective transport; The Cross-mesh topology demonstrates competitive early-stage convergence and balanced accuracy across all four scenarios, with its full-connectivity entanglement yielding robust performance for fully coupled pressure-saturation two-phase flow and emerging as a reliable choice for multi-physics cross-coupling problems.

Numerical experiments on four typical reservoir seepage models confirm that QCPINNs can reliably capture all core reservoir flow characteristics, including pressure distribution in heterogeneous media, steep

saturation fronts in waterflooding, adsorption-retarded concentration migration in compositional flow, and the coupled evolution of pressure and saturation in heterogeneous two-phase oil-water seepage with exponential permeability variation. Quantitative error analysis indicates that all QCPINN topologies yield low magnitude errors, with QCPINNs outperforming classical PINNs by up to 20 times in pressure prediction and 5 times in saturation prediction for coupled two-phase flow; the Alternate topology even achieves better front discontinuity capture than high-precision numerical solutions with ultra-dense grids, validating the method's accuracy, stability and superiority over classical methods.

This work extends the QCPINN from fundamental physical PDEs to the full spectrum of core seepage problems encountered in industrial reservoir engineering, offering a new technical pathway for the development of high-efficiency reservoir simulators and machine learning surrogate models. The proposed QCPINN framework can be readily integrated with existing PINN improvement strategies, which provides a solid basis for future extensions toward real-world reservoirs involving strong heterogeneity, complex fracture networks, coupled multiphase/multicomponent flow, transient displacement, and complex well patterns, thereby enhancing practical applicability and motivating further research.

In summary, this work verifies that QCPINNs offer a breakthrough solution to the limitations of classical numerical methods and PINNs in complex reservoir simulation. It lays a solid foundation for the industrial application of quantum computing in oil and gas field development, and provides clear guidance for the problem-adaptive design of quantum circuit topologies in reservoir engineering and other computational geoscience fields.

6 Acknowledgements

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